

Practical Regularized Quasi-Newton Methods with Inexact Function Values

Hiroki Hamaguchi  · Naoki Marumo  ·
Akiko Takeda 

Received: date / Accepted: date

Abstract Many practical optimization problems involve objective function values that are corrupted by bounded, nonvanishing numerical errors. We propose a noise-tolerant regularized quasi-Newton method that uses a relaxed Armijo line search when observed function values certify progress and switches to OFFO regularization otherwise. With exact gradients and uniform spectral bounds on the quasi-Newton matrices, we prove an order-optimal $\mathcal{O}(1/\varepsilon^2)$ complexity bound under bounded relative-or-absolute errors in function values. Although the method uses persistently noisy function values when they certify progress, it retains the optimal worst-case dependence on the stationarity tolerance for the noise-free L -smooth nonconvex setting. The exact-gradient qualification is essential to the stated theorem. Experiments with perturbed gradients are reported separately as deliberately adverse robustness tests beyond its scope. Experiments on CUTEst include function-value-only noise, joint function-and-gradient noise, and 64-, 32-, and 16-bit arithmetic. They evaluate both robustness and the sensitivity to the supplied function-error bound.

Keywords Regularized Quasi-Newton Method · Numerical Stability · Numerical Error · AdaGrad-Norm · Objective-Function-Free Optimization

Mathematics Subject Classification (2020) 90C53 · 90C06 · 65K05 · 49M15

H. Hamaguchi
Graduate School of Information Science and Technology, The University of Tokyo, Tokyo, Japan
E-mail: hirokihamaguchi0331@gmail.com

N. Marumo
Graduate School of Information Science and Technology, The University of Tokyo, Tokyo, Japan

A. Takeda
Graduate School of Information Science and Technology, The University of Tokyo, Tokyo, Japan
Center for Advanced Intelligence Project, RIKEN, Tokyo, Japan

1 Introduction

We consider the unconstrained optimization problem

$$\underset{x \in \mathbb{R}^n}{\text{minimize}} \quad f(x), \quad (1)$$

where $f: \mathbb{R}^n \rightarrow \mathbb{R}$ is a continuously differentiable nonconvex function, and $x \in \mathbb{R}^n$ is the optimization variable. We focus on settings where objective function evaluations are contaminated by bounded and non-diminishing numerical noise [41]. Specifically, only inexact function values $\bar{f}(x)$ are available for optimization. Such inaccuracies are unavoidable in many realistic scenarios, including finite-precision arithmetic, simulation-based evaluations, and stochastic approximations [5, 32, 40]. When function values are contaminated by numerical noise, standard optimization algorithms may lose their theoretical guarantees, behave erratically, or terminate prematurely with unreliable solutions.

In the absence of noise and with accurate function evaluations, quasi-Newton methods, in particular the Broyden–Fletcher–Goldfarb–Shanno (BFGS) method and its limited-memory variant (L-BFGS), are among the most efficient and widely used algorithms for solving problem (1) [28, 34]. Their success is largely attributed to fast local convergence and scalability. Standard implementations [35, 44] rely on line search procedures that enforce the Wolfe conditions to guarantee sufficient descent and the positive definiteness of the Hessian approximation.

In the presence of numerical noise, these line search conditions may become unreliable. Differences in function values or directional derivatives can be dominated by numerical errors, leading to unstable step sizes, ill-conditioned Hessian approximations, or abnormal termination [36, 40]. From a computational perspective, improving the robustness of quasi-Newton methods in noisy environments is therefore crucial. In particular, it is desirable to design algorithms that (i) retain computational efficiency when function evaluations are accurate, (ii) remain stable when function values are unreliable, and (iii) guarantee convergence to first-order stationary points under mild assumptions.

In this work, we propose a noise-tolerant regularized quasi-Newton method that addresses these challenges. Our approach combines a relaxed Armijo line search, quadratic regularization [25, 43], and an OFFO-inspired strategy [19, 21]. The closest regularized L-BFGS result is Theorem 1 of Kanzow and Steck [25]. Under bounded quasi-Newton matrices and a lower-bounded objective, it proves that the gradient norm has limit inferior zero. It assumes reliable function values and gives no complexity bound. Our result instead gives the order-optimal $\mathcal{O}(1/\varepsilon^2)$ bound for smooth nonconvex optimization under persistent bounded relative-or-absolute function errors [6]. Rather than merely relaxing the line search, our method uses noisy function values only while they certify a reference decrease and otherwise switches to an objective-function-free regularization scale. It thereby retains useful line-search behavior without making continued progress depend on unreliable comparisons.

OFFO methods avoid unreliable function values altogether [19, 21], whereas our method uses them selectively. The noise-tolerant quasi-Newton analysis of Shi et al. [40] permits both function and gradient errors but assumes strong convexity

Algorithm	Uses function values	Tolerant to Function noise	Gradient noise allowed	Nonconvex	Rate / complexity (target)
Line / SciPy [28, 35, 44]	✓	–	–	✓	R-linear objective gap
L-BFGS-type line search [8]	✓	–	–	✓	$\mathcal{O}(\varepsilon^{-2})$ evaluations to $\ \nabla f\ \leq \varepsilon$
Reg [25]	✓	–	–	✓	–
NTQN [40, 47]	✓	✓	✓	–	R-linear objective gap to the noise floor
OFFO [19]	–	✓	–	✓	$\mathcal{O}(\varepsilon^{-2})$ to $\ \nabla f\ \leq \varepsilon$
Ours Algorithm 1	✓	✓	–*	✓	$\mathcal{O}(\varepsilon^{-2})$ to $\ \nabla f\ \leq \varepsilon$

Table 1: Quasi-Newton-focused comparison of the proposed method and its baselines. OFFO is included as the first-order source of the fallback scaling. A ✓ denotes theoretical coverage and “–” its absence. The asterisk marks empirical robustness outside our gradient-noise theory.

and converges linearly only to a noise-dependent neighborhood. Its stationarity radius contains a function-error contribution proportional to $\sqrt{M\varepsilon_f}$ and a gradient-error contribution proportional to ε_g , so keeping the former below the latter requires $\varepsilon_f = \mathcal{O}(\varepsilon_g^2/M)$ up to problem-dependent constants. In particular, even when $\varepsilon_g = 0$, their bound retains a threshold proportional to $\sqrt{M\varepsilon_f}$, whereas a fixed $0 \leq \varepsilon_f < 1$ changes the constants in our bound but imposes no stationarity floor. This distinction follows from our adaptively recomputed error allowance and the K^0/K^+ switch, which stops relying on function comparisons when they cease to certify progress. Shi et al. cover bounded gradient errors and protect curvature pairs by lengthening, whereas our arbitrary-accuracy and complexity guarantees require exact gradients. Our relaxed Armijo condition is also directly related to the noise-tolerant relaxation in Section 4 of Shi et al. [40]. When the relevant function values are small enough that the constant branch of the relative-or-absolute error model is active, our adaptive allowance reduces to $\Delta_k = 2\varepsilon_f/(1 - \varepsilon_f)$ and directly parallels their constant error allowance.

The rates in Table 1 have different targets and assumptions. For line-search L-BFGS with exact evaluations, Theorem 7.1 of Liu and Nocedal [28] proves R-linear convergence of the objective gap under uniform convexity and bounded initial scaling matrices. Theorem 3.4 of Cartis et al. [8] gives an $\mathcal{O}(1/\varepsilon^2)$ function- and gradient-evaluation bound for smooth nonconvex optimization when the line-search direction is uniformly gradient-related. This result covers L-BFGS-type variable-metric directions when their matrices are uniformly well conditioned, but it assumes exact function values and gradients. For NTQN, Theorem 3.5 of Shi et al. [40] proves $\phi(x_k) - \phi^* \leq \rho^k(\phi(x_0) - \phi^*) + 2\varepsilon_f$, where $0 < \rho < 1$, until the iterates enter a noise-dependent neighborhood. Thus NTQN has an order-level linear rate under strong convexity, but does not guarantee an arbitrary smaller stationarity tolerance. By contrast, OFFO and our method have $\mathcal{O}(1/\varepsilon^2)$ worst-case complexity for first-order station-

arity on smooth nonconvex objectives, and our result additionally permits persistent bounded function errors.

When the observed function value certifies sufficient decrease, the method uses an unregularized quasi-Newton step. Otherwise, an AdaGrad-Norm-type accumulator controls the regularization over iterations in K^+ . Under exact gradients and bounded relative-or-absolute function errors, the method reaches an ε -first-order stationary point in the order-optimal $\mathcal{O}(1/\varepsilon^2)$ iterations. The CUTEst study reports problem dimensions, stopping rules, sensitivity to the supplied error estimate, and ablations of the hybrid mechanism. It separates theory-aligned function-noise experiments from joint function-and-gradient perturbations, which are deliberately adverse stress tests rather than evidence for the theorem.

The remainder of this paper is organized as follows. In section 2, we review relevant background and introduce the problem setting. In section 3, we present the proposed algorithm. In section 4, we establish a global convergence property of the proposed algorithm. In section 5, we discuss practical choices of algorithmic variables and parameters. In section 6, we report numerical results on the CUTEst benchmark collection. Finally, in section 7, we conclude the paper with a summary and future research directions.

Our implementation is publicly available on GitHub¹.

2 Preliminaries and Problem Setting

We begin by reviewing relevant background and formalizing the problem setting.

2.1 Regularized Quasi-Newton Method

A regularized quasi-Newton method is a variant of quasi-Newton methods that does not rely heavily on line-search techniques. This property is important for optimization in noisy evaluation environments. There has been renewed interest in regularized Newton-type methods, including cubic-regularization schemes [33, 37, 45], and we focus on quadratic regularization [11, 25, 27, 39, 42, 43, 50] for simplicity of analysis and implementation.

For $g_k := \nabla f(x_k)$ and an approximate Hessian $B_k \in \mathbb{R}^{n \times n}$, regularized quasi-Newton methods compute the step as

$$x_{k+1} = x_k + d_k, \quad d_k = -(B_k + \mu_k I)^{-1} g_k,$$

with a regularization parameter $\mu_k \geq 0$. Combining regularization with line search is also possible and has been explored [42]. Several strategies for choosing μ_k in nonconvex optimization have been proposed [39, 42, 43], and one of them is given by

$$\mu_k = c_1 \max(0, -\lambda_{\min}(B_k)) + c_2 \|\nabla f(x_k)\|^\delta,$$

¹ <https://github.com/HirokiHamaguchi/qnlab>

where $c_1 > 1$, $c_2 > 0$, $\delta \geq 0$, and $\lambda_{\min}(B_k)$ denotes the smallest eigenvalue of B_k . This choice guarantees that $B_k + \mu_k I$ is positive definite, ensuring that d_k is a descent direction. We adapt such regularization strategies to noisy evaluation environments.

2.2 Objective-Function-Free Optimization

Objective-Function-Free Optimization (OFFO) refers to a class of methods that avoid explicit evaluations of the objective function and rely solely on gradient information [19, 21]. Such methods are particularly attractive when function values are unreliable due to noise. Similar adaptive trust-region approaches are also discussed [18].

Prominent examples of OFFO include adaptive gradient methods such as AdaGrad [13] and AdaGrad-Norm [46]. Their update rules are

$$\begin{aligned} \text{(AdaGrad)} \quad x_{k+1}^{(i)} &= x_k^{(i)} - \frac{1}{\theta \sqrt{\varsigma + \sum_{j=0}^k (g_j^{(i)})^2}} g_k^{(i)}, \quad (i = 1, 2, \dots, n) \\ \text{(AdaGrad-Norm)} \quad x_{k+1} &= x_k - \frac{1}{\theta \sqrt{\varsigma + \sum_{j=0}^k \|g_j\|^2}} g_k \end{aligned}$$

respectively, where $\varsigma > 0$ and $\theta > 0$ are algorithmic parameters, and $x^{(i)}$ denotes the i -th component of a vector x .

2.3 Problem Setting

We consider optimization problems in which exact objective function values are unavailable, following the recent literature on error-aware optimization [2, 7, 9, 20, 38, 41]. When both objective values and gradients are heavily corrupted by noise, reliable optimization cannot be guaranteed. Thus, meaningful optimization requires that at least one source of information be sufficiently accurate. This work focuses on problems arising from finite-precision computations or inexact evaluations, where objective function evaluations are subject to non-negligible errors while gradient information can be computed with relatively high accuracy. Settings dominated by gradient errors, such as stochastic approximation and inexact-gradient methods [3, 40, 47], are beyond the scope of this work.

In the following, we formalize the assumptions on function value and gradient evaluations. We assume that only an error-contaminated function value $\bar{f}(x)$ is accessible, satisfying the hybrid absolute-relative error model [23, 24]:

$$|f(x) - \bar{f}(x)| \leq \varepsilon_f \max(1, |f(x)|), \quad (2)$$

where $0 \leq \varepsilon_f < 1$ is an error rate. The error is predominantly relative when $|f(x)|$ is large and effectively absolute when $|f(x)|$ is small, capturing the typical behavior of rounding errors in IEEE 754 floating-point arithmetic [23]. The value of ε_f serves as a safety margin for errors in the computed function values. Underestimating ε_f

can violate eq. (2) and cause the method to treat unreliable function-value comparisons as informative. By contrast, choosing a larger value preserves the analysis as long as $\varepsilon_f < 1$ and the error bound remains valid, although it moves the method earlier or more frequently into the more conservative OFFO-type regime K^+ . Because a function evaluation generally involves many floating-point operations whose errors can accumulate, we choose a precision-dependent multiple of machine epsilon as an aggregate error allowance rather than identifying ε_f with the error from a single rounding operation. Specifically, our choices are guided by numerical stopping rules such as the default SciPy L-BFGS-B function tolerance, which uses 10^7 times machine epsilon [44]. The values used at each precision and a sensitivity study with under- and overestimated values are reported in section 6.

We next specify the assumption on gradient evaluations. In practice, termination is typically based on a gradient tolerance $\varepsilon_{\text{gtol}}$, and $\|g(x)\| \leq \varepsilon_{\text{gtol}}$ is used as a stopping criterion. Thus, before termination, $\|g(x)\| > \varepsilon_{\text{gtol}}$. If the gradient error $\|g(x) - \nabla f(x)\|$ were comparable to or larger than $\varepsilon_{\text{gtol}}$, this stopping criterion would not reliably indicate stationarity even when $\nabla f(x) = 0$. Meaningful termination therefore requires the gradient error to be sufficiently small relative to $\varepsilon_{\text{gtol}}$. In particular, before termination, we have in mind the regime $\|g(x) - \nabla f(x)\| \ll \varepsilon_{\text{gtol}} < \|g(x)\|$. Under this regime, $g(x)$ is a sufficiently accurate approximation of $\nabla f(x)$ for the required accuracy. For simplicity, the convergence analysis adopts the following exact-gradient assumption.

Assumption 1 (Exact gradient) *The algorithm has access to a gradient oracle g such that, for every evaluated point x ,*

$$g(x) = \nabla f(x). \quad (3)$$

This assumption is a modeling simplification for the analysis rather than a claim that practical gradients are exact. The proposed method does not use Wolfe conditions and therefore does not require gradient evaluations at trial points to verify such conditions. The function-value-only experiments satisfy Assumption 1, whereas the joint-noise experiments deliberately violate it and lie outside the theory. The latter experiments are included to assess the empirical robustness of the proposed method, not to validate or extend the convergence result.

3 Proposed Algorithm

We propose a new algorithm for the problem setting stated in section 2.3. We draw inspiration from the OFFO framework to design robust regularization and scaling strategies. Unlike purely OFFO methods, our algorithm exploits function values when they are numerically reliable and smoothly switches to gradient-based control otherwise. A regularized quasi-Newton direction alone does not specify how to update the regularization scale once comparisons of noisy function values become uninformative, whereas the cumulative gradient scale provides such an update without further objective comparisons. Using that scale at every iteration would instead discard useful function information and may be unnecessarily conservative, which motivates the

K^0/K^+ switch. This hybrid behavior is intended to balance stability under unreliable function values with the efficiency of unregularized quasi-Newton steps when function information is useful. The pseudocode of our algorithm is given in Algorithm 1. The minimum of the empty set is defined as $+\infty$ in line 3. Algorithm 2 is an auxiliary line search algorithm called in Algorithm 1. In the following, we explain the three main components of Algorithm 1.

Algorithm 1: Noise Tolerant Regularized Quasi-Newton Method

Input: $x_0, 0 < k_{\max}, 0 \leq \varepsilon_f < 1, 0 < \theta_{\min} \leq \theta_{\max}$ and $0 < \varsigma < 1$

- 1 $k \leftarrow 0, g_0 \leftarrow g(x_0), K^0 \leftarrow \emptyset, K^+ \leftarrow \emptyset$
- 2 **for** $k = 0, 1, 2, \dots, k_{\max} - 1$ **do**
- 3 **if** $\min_{j \in K^0, j < k} (\bar{f}(x_j) - \Delta_j) \geq \bar{f}(x_k)$ **then**
- 4 $K^0 \leftarrow K^0 \cup \{k\}$
- 5 $\mu_k \leftarrow 0$
- 6 **else**
- 7 $K^+ \leftarrow K^+ \cup \{k\}$
- 8 $\mu_k \leftarrow \theta_k \sqrt{\varsigma + \sum_{j \in K^+, j \leq k} \|g_j\|^2}$ with $\theta_k \in [\theta_{\min}, \theta_{\max}]$ (see section 5.3).
- 9 $d_k \leftarrow -(B_k + \mu_k I)^{-1} g_k$ with B_k (see section 5.1).
- 10 Find α_k and Δ_k satisfying eq. (4) by Algorithm 2.
- 11 $x_{k+1} \leftarrow x_k + \alpha_k d_k, g_{k+1} \leftarrow g(x_{k+1})$
- 12 **end**

Algorithm 2: Backtracking Line Search with Relaxed Armijo Condition

Input: $x_k \in \mathbb{R}^n, d_k \in \mathbb{R}^n, g_k \in \mathbb{R}^n, 0 < c < 1$ and $0 < \beta_{\min} < \beta_{\max} < 1$

- 1 $\alpha_k \leftarrow 1$
- 2 $\Delta_k \leftarrow \frac{2\varepsilon_f}{1-\varepsilon_f} \max(1, \bar{f}(x_k), -\bar{f}(x_k + \alpha_k d_k))$
- 3 **while** $\bar{f}(x_k) + c\alpha_k g_k^\top d_k + \Delta_k < \bar{f}(x_k + \alpha_k d_k)$ **do**
- 4 $\alpha_k \leftarrow \beta \alpha_k$ with $\beta \in [\beta_{\min}, \beta_{\max}]$ (see section 5.2).
- 5 $\Delta_k \leftarrow \frac{2\varepsilon_f}{1-\varepsilon_f} \max(1, \bar{f}(x_k), -\bar{f}(x_k + \alpha_k d_k))$
- 6 **end**
- 7 **return** α_k

Computation of the Regularized Quasi-Newton Direction (line 9): The first component is the computation of the regularized quasi-Newton direction d_k stated in section 2.1. Given $g_k = g(x_k) = \nabla f(x_k)$, an approximate Hessian B_k , and a regularization parameter μ_k , we compute the search direction $d_k = -(B_k + \mu_k I)^{-1} g_k$. A practical choice of B_k is discussed in section 5.1.

Relaxed Armijo Condition with an Error-Absorbing Term (line 10): The second component is the computation of the step size α_k using a relaxed Armijo condition, which is defined as follows:

$$\begin{cases} \bar{f}(x_k) + c\alpha_k g_k^\top d_k + \Delta_k \geq \bar{f}(x_k + \alpha_k d_k), \\ \Delta_k = \frac{2\varepsilon_f}{1 - \varepsilon_f} \max(1, \bar{f}(x_k), -\bar{f}(x_k + \alpha_k d_k)), \end{cases} \quad (4)$$

where Δ_k is the error-absorbing term that is recomputed for each α_k . This relaxation guarantees the existence of $\alpha_k \in (0, 1]$ even in the presence of errors in $\bar{f}$. When d_k is a descent direction ($g_k^\top d_k < 0$), eq. (4) ensures that a temporary increase of $\bar{f}$ is at most Δ_k . Since Δ_k depends on $-\bar{f}(x_k + \alpha_k d_k)$ but not on $\bar{f}(x_k + \alpha_k d_k)$, Δ_k does not increase when $\bar{f}(x_k + \alpha_k d_k)$ becomes large. This prevents unbounded growth of Δ_k .

Adaptive Update of the Regularization Parameter (lines 3–8): The third component is the adaptive update of the regularization parameter μ_k . A larger μ_k yields a more conservative step, while a smaller μ_k allows a more aggressive quasi-Newton step, especially when $\mu_k = 0$. In practical settings, $\mu_k = 0$ is preferred, since this allows us to fully exploit the quasi-Newton approximation and typically leads to practically faster convergence. Let us partition the iteration indices $K := \{0, 1, 2, \dots, k_{\max} - 1\}$ into two sets:

$$K^0 := \{k \in K \mid \mu_k = 0\}, \quad K^+ := \{k \in K \mid \mu_k > 0\}. \quad (5)$$

We set $\mu_k = 0$ if and only if the following condition is satisfied:

$$\min_{j \in K^0, j < k} (\bar{f}(x_j) - \Delta_j) \geq \bar{f}(x_k). \quad (6)$$

Note that $k = 0$ satisfies eq. (6) since K^0 is empty. This condition restricts $\mu_k = 0$ to iterations for which a sufficient decrease in $\bar{f}$ has been observed. Thus, although $\bar{f}$ can temporarily increase, every iteration in K^0 must improve the previous error-adjusted reference value. When this test fails, the method instead enters K^+ and uses the OFFO-based regularization update below. Otherwise, to ensure convergence to first-order stationary points, we apply an OFFO-based update

$$\mu_k = \theta_k \sqrt{\varsigma + \sum_{j \in K^+, j \leq k} \|g_j\|^2}, \quad (7)$$

as described in section 2.2. Since OFFO schemes do not rely on function values, this update is robust to numerical errors in $\bar{f}$.

4 Theoretical Analysis

In this section, we establish the global convergence properties of Algorithm 1 under the problem setting in section 2.3, namely, eqs. (2) and (3).

4.1 Assumptions

In addition to the exact-gradient assumption stated in Assumption 1, we make the following three structural assumptions. The first assumption ensures that the objective function is bounded below.

Assumption 2 *The function f is bounded below, meaning that for all $x \in \mathbb{R}^n$,*

$$f(x) \geq f^* := \inf_{x \in \mathbb{R}^n} f(x) > -\infty.$$

By the triangle inequality and eq. (2), we have $\bar{f}(x) \geq f(x) - \varepsilon_f \max(1, |f(x)|)$. Since the function $g(t) = t - \varepsilon_f \max(1, |t|)$ is non-decreasing for $0 \leq \varepsilon_f < 1$, it follows that $\inf_x g(f(x)) = g(f^*)$. Consequently, we can also lower bound $\bar{f}$ as

$$\bar{f}(x) \geq \bar{f}^* := \inf_{x \in \mathbb{R}^n} (f(x) - \varepsilon_f \max(1, |f(x)|)) = f^* - \varepsilon_f \max(1, |f^*|) > -\infty. \quad (8)$$

The second assumption ensures smoothness of the objective function.

Assumption 3 *The function $f: \mathbb{R}^n \rightarrow \mathbb{R}$ is L -smooth, meaning that there exists a constant $L > 0$ such that for all $x, y \in \mathbb{R}^n$,*

$$f(y) \leq f(x) + \nabla f(x)^\top (y - x) + \frac{L}{2} \|y - x\|^2.$$

The third assumption requires the matrices $\{B_k\}_{k=0}^{k_{\max}-1}$ to be uniformly positive definite and bounded.

Assumption 4 *There exist constants $0 < m \leq M$ such that all B_k satisfy*

$$mI \preceq B_k \preceq MI.$$

As we will discuss in section 5, the standard BFGS update with a slight modification guarantees Assumption 4 in practice. By Assumption 4, we can derive useful bounds involving g_k and d_k . Using $d_k = -(B_k + \mu_k I)^{-1} g_k$ (line 9) and Assumption 4 asserting $(m + \mu_k)I \preceq B_k + \mu_k I \preceq (M + \mu_k)I$, we obtain the following bounds:

$$-g_k^\top d_k = d_k^\top (B_k + \mu_k I) d_k \geq (m + \mu_k) \|d_k\|^2 \geq m \|d_k\|^2, \quad (9)$$

$$-g_k^\top d_k = g_k^\top (B_k + \mu_k I)^{-1} g_k \geq \frac{1}{M + \mu_k} \|g_k\|^2, \quad (10)$$

$$\|d_k\|^2 = \|(B_k + \mu_k I)^{-1} g_k\|^2 \leq \frac{1}{(m + \mu_k)^2} \|g_k\|^2. \quad (11)$$

4.2 Error Bound on Function Evaluations

We first establish useful properties of the error-absorbing term Δ_k defined in eq. (4). In the following, we define the actual and observed function decreases as

$$\delta_k := f(x_k) - f(x_{k+1}), \quad (12)$$

$$\bar{\delta}_k := \bar{f}(x_k) - \bar{f}(x_{k+1}). \quad (13)$$

The first property states that the difference between the actual decrease δ_k and the observed decrease $\bar{\delta}_k$ can be bounded by Δ_k .

Lemma 1 *If x_k and x_{k+1} satisfy $f(x_{k+1}) \leq f(x_k)$, it holds that*

$$\delta_k \leq \bar{\delta}_k + \Delta_k.$$

Proof We start from general observations. We first prove that, for all $x \in \mathbb{R}^n$,

$$\begin{cases} \max(1, +f(x)) \leq \frac{1}{1-\varepsilon_f} \max(1, +\bar{f}(x)), \\ \max(1, -f(x)) \leq \frac{1}{1-\varepsilon_f} \max(1, -\bar{f}(x)). \end{cases} \quad (14)$$

We prove the inequality with the plus sign. If $+f(x) \leq 1$, then $\max(1, +f(x)) = 1 \leq \frac{1}{1-\varepsilon_f} \leq \frac{1}{1-\varepsilon_f} \max(1, +\bar{f}(x))$. If $+f(x) > 1$, eq. (2) implies

$$|f(x) - \bar{f}(x)| \leq \varepsilon_f \max(1, |f(x)|) = +\varepsilon_f f(x) \quad (15)$$

and thus $f(x) - \bar{f}(x) \leq +\varepsilon_f f(x)$. Hence,

$$\max(1, +f(x)) = +f(x) \leq +\frac{1}{1-\varepsilon_f} \bar{f}(x) \leq \frac{1}{1-\varepsilon_f} \max(1, +\bar{f}(x)).$$

The case with the negative sign can be shown analogously, using $-f(x) + \bar{f}(x) \leq -\varepsilon_f f(x)$, which follows from eq. (15). This completes the proof of eq. (14).

We also have the following general bound for all $k \in K$:

$$\begin{aligned} |\delta_k - \bar{\delta}_k| &= |(f(x_k) - \bar{f}(x_k)) - (f(x_{k+1}) - \bar{f}(x_{k+1}))| && \text{(rearrangement)} \\ &\leq |f(x_k) - \bar{f}(x_k)| + |f(x_{k+1}) - \bar{f}(x_{k+1})| && \text{(triangle inequality)} \\ &\leq \varepsilon_f \max(1, |f(x_k)|) + \varepsilon_f \max(1, |f(x_{k+1})|) && \text{(eq. (2))} \\ &\leq 2\varepsilon_f \max(1, |f(x_k)|, |f(x_{k+1})|). && \text{(max property)} \end{aligned} \quad (16)$$

With these preparations, we now prove the lemma. From the condition $f(x_{k+1}) \leq f(x_k)$, eq. (16) can be further bounded as

$$\begin{aligned} 2\varepsilon_f \max(1, |f(x_k)|, |f(x_{k+1})|) &= 2\varepsilon_f \max(1, f(x_k), -f(x_{k+1})) && (f(x_{k+1}) \leq f(x_k)) \\ &\leq \frac{2\varepsilon_f}{1-\varepsilon_f} \max(1, \bar{f}(x_k), -\bar{f}(x_{k+1})) && \text{(eq. (14))} \\ &= \Delta_k, && \text{(eq. (4))} \end{aligned}$$

which yields the desired inequality with $\delta_k - \bar{\delta}_k \leq |\delta_k - \bar{\delta}_k| \leq \Delta_k$. $\square$

We also derive the following lemma for later use.

Lemma 2 *Under Assumption 4, it holds that*

$$\bar{\delta}_k \leq \delta_k + \frac{1 + \varepsilon_f}{1 - \varepsilon_f} \Delta_k.$$

Proof Since $-g_k^\top d_k \geq 0$ from Assumption 4 and eq. (9), the relaxed Armijo condition (eq. (4)) implies

$$\bar{f}(x_k) - \bar{f}(x_{k+1}) \geq -c\alpha_k g_k^\top d_k - \Delta_k \geq -\Delta_k. \quad (17)$$

Thus, using the general results in eq. (16) and $\Delta_k \geq 0$, we can further bound as

$$\begin{aligned} |\delta_k - \bar{\delta}_k| &\leq 2\varepsilon_f \max(1, |f(x_k)|, |f(x_{k+1})|) && \text{(eq. (16))} \\ &\leq \frac{2\varepsilon_f}{1 - \varepsilon_f} \max(1, \bar{f}(x_k), -\bar{f}(x_k), \bar{f}(x_{k+1}), -\bar{f}(x_{k+1})) && \text{(eq. (14))} \\ &\leq \frac{2\varepsilon_f}{1 - \varepsilon_f} \max(1, \bar{f}(x_k) + \Delta_k, -\bar{f}(x_{k+1}) + \Delta_k) && \text{(eq. (17))} \\ &\leq \frac{2\varepsilon_f}{1 - \varepsilon_f} (\max(1, \bar{f}(x_k), -\bar{f}(x_{k+1})) + \Delta_k) \\ &= \left(1 + \frac{2\varepsilon_f}{1 - \varepsilon_f}\right) \Delta_k = \frac{1 + \varepsilon_f}{1 - \varepsilon_f} \Delta_k, && \text{(eq. (4))} \end{aligned}$$

which yields the desired inequality with $\bar{\delta}_k - \delta_k \leq |\delta_k - \bar{\delta}_k| \leq \frac{1 + \varepsilon_f}{1 - \varepsilon_f} \Delta_k$. $\square$

4.3 Backtracking Stage

We next analyze the line search in Algorithm 2. Recall that the parameter c in Algorithm 2 satisfies $0 < c < 1$ and $0 < m$ from Assumption 4.

Lemma 3 *Under Assumptions 3 and 4, the backtracking procedure in Algorithm 2 terminates whenever the step size α_k satisfies*

$$\alpha_k \leq \frac{2(1-c)m}{L}. \quad (18)$$

Proof Under the L -smoothness of f (Assumption 3), we always have

$$\begin{aligned} f(x_k) - f(x_k + \alpha_k d_k) &\geq -\alpha_k g_k^\top d_k - \frac{L}{2} \alpha_k^2 \|d_k\|^2 \\ &= -c\alpha_k g_k^\top d_k + \alpha_k \left(-(1-c)g_k^\top d_k - \frac{L\alpha_k}{2} \|d_k\|^2 \right). \end{aligned} \quad (19)$$

From Assumption 4 and eq. (9), we have $-g_k^\top d_k \geq m\|d_k\|^2$. Under eq. (18), we have

$$-(1-c)g_k^\top d_k - \frac{L\alpha_k}{2} \|d_k\|^2 \geq \left((1-c)m - \frac{L\alpha_k}{2} \right) \|d_k\|^2 \geq 0. \quad (20)$$

Combining eqs. (19) and (20) yields

$$f(x_k) - f(x_k + \alpha_k d_k) \geq -c\alpha_k g_k^\top d_k.$$

Since $-g_k^\top d_k \geq 0$ from Assumption 4 and eq. (9), the condition of Lemma 1 is satisfied with $x_{k+1} = x_k + \alpha_k d_k$. Thus, we obtain $\delta_k \leq \bar{\delta}_k + \Delta_k$, i.e.,

$$-c\alpha_k g_k^\top d_k \leq f(x_k) - f(x_k + \alpha_k d_k) \leq \bar{f}(x_k) - \bar{f}(x_k + \alpha_k d_k) + \Delta_k.$$

This inequality means that the relaxed Armijo condition (eq. (4)) is satisfied and thus the backtracking procedure terminates. This completes the proof. $\square$

Now, let us analyze Algorithm 2 using Lemma 3. Define

$$\bar{\alpha} := \min\left(\beta_{\min} \frac{2(1-c)m}{L}, 1\right) > 0.$$

The following states the key property of the backtracking procedure.

Proposition 1 *Under Assumptions 3 and 4, the backtracking line search in Algorithm 2 terminates after finitely many rejections, and the step size α_k satisfies*

$$\alpha_k \geq \bar{\alpha}.$$

Proof Line 4 of Algorithm 2 multiplies α_k by β at each rejection, where $0 < \beta_{\min} \leq \beta \leq \beta_{\max} < 1$. For the number of backtracking rejections r , we have $\alpha_k \leq \beta_{\max}^r$. Thus, r is finite since the backtracking procedure terminates for sufficiently small α_k by Lemma 3. If $r = 0$, $\alpha_k = 1 \geq \bar{\alpha}$. Otherwise, $\alpha_k \geq \beta_{\min} \alpha'_k$, where α'_k is the last rejected step size. By Lemma 3, we have $\alpha'_k > \frac{2(1-c)m}{L}$, and thus

$$\alpha_k \geq \beta_{\min} \alpha'_k > \beta_{\min} \frac{2(1-c)m}{L} \geq \bar{\alpha}.$$

This completes the proof. $\square$

4.4 Bound for the case $\mu_k = 0$

In this subsection, we lower bound the sum of actual function decreases δ_k over $k \in K^0$. Recall that K^0 denotes the set of iteration indices k such that $\mu_k = 0$, as in eq. (5). To this end, we establish that the sum of Δ_k over $k \in K^0$ is upper bounded, where Δ_k is the error-absorbing term in the relaxed Armijo condition (eq. (4)). We introduce the following constant $\Delta_{\max}$, which serves as an upper bound for Δ_k with $k \in K^0$.

$$\Delta_{\max} := \frac{2\varepsilon_f}{1-\varepsilon_f} \max(1, \bar{f}(x_0), -\bar{f}^*).$$

Lemma 4 *Under Assumptions 2 to 4, the sum of Δ_k over $k \in K^0$ satisfies*

$$\sum_{k \in K^0} \Delta_k \leq \bar{f}(x_0) - \bar{f}^* + \Delta_{\max}.$$

Proof Let $(k_i)_{i=0}^\ell$ with $\ell \geq 0$ be the elements of K^0 in increasing order. We have $\bar{f}(x_{k_\ell}) \leq \bar{f}(x_0)$ from eq. (6) with $k = k_\ell$, where the case $\ell = 0$ follows from $k_0 = 0$. Combining this with Assumption 2, we have $\Delta_{k_\ell} \leq \Delta_{\max}$. For all $i < \ell$, eq. (6) with $j = k_i$ and $k = k_{i+1}$ yields

$$\Delta_{k_i} \leq \bar{f}(x_{k_i}) - \bar{f}(x_{k_{i+1}}), \quad (21)$$

and summing Δ_k over $k \in K^0$ gives

$$\begin{aligned} \sum_{k \in K^0} \Delta_k &\leq \sum_{i=0}^{\ell-1} (\bar{f}(x_{k_i}) - \bar{f}(x_{k_{i+1}})) + \Delta_{k_\ell} && \text{(eq. (21))} \\ &= \bar{f}(x_0) - \bar{f}(x_{k_\ell}) + \Delta_{k_\ell} && \text{(telescoping sum)} \\ &\leq \bar{f}(x_0) - \bar{f}^* + \Delta_{\max}. && (-\bar{f}(x_{k_\ell}) \leq -\bar{f}^* \text{ and } \Delta_{k_\ell} \leq \Delta_{\max}) \end{aligned}$$

This completes the proof. $\square$

We now bound the sum of $\delta_k = f(x_k) - f(x_{k+1})$ over $k \in K^0$ with $\|g_k\|^2$ as follows.

Lemma 5 *Under Assumptions 2 to 4, the sum of δ_k over $k \in K^0$ satisfies*

$$\sum_{k \in K^0} \delta_k \geq \frac{c\bar{\alpha}}{M} \sum_{k \in K^0} \|g_k\|^2 - \frac{2}{1 - \varepsilon_f} (\bar{f}(x_0) - \bar{f}^* + \Delta_{\max}).$$

Proof The relaxed Armijo condition (eq. (4)) at iteration $k \in K^0$ yields

$$\begin{aligned} \bar{\delta}_k + \Delta_k &= \bar{f}(x_k) - \bar{f}(x_{k+1}) + \Delta_k && \text{(eq. (13))} \\ &\geq -c\alpha_k g_k^\top d_k && \text{(eq. (4))} \\ &\geq \frac{c\bar{\alpha}}{M + \mu_k} \|g_k\|^2 && \text{(Proposition 1 and eq. (10))} \\ &= \frac{c\bar{\alpha}}{M} \|g_k\|^2. && (\mu_k = 0 \text{ since } k \in K^0) \end{aligned} \quad (22)$$

Using Lemma 2, we also have

$$\bar{\delta}_k + \Delta_k \leq \delta_k + \frac{1 + \varepsilon_f}{1 - \varepsilon_f} \Delta_k + \Delta_k = \delta_k + \frac{2}{1 - \varepsilon_f} \Delta_k. \quad (23)$$

Then, combining eqs. (22) and (23) and summing over $k \in K^0$ gives

$$\sum_{k \in K^0} \frac{c\bar{\alpha}}{M} \|g_k\|^2 \leq \sum_{k \in K^0} \delta_k + \frac{2}{1 - \varepsilon_f} \sum_{k \in K^0} \Delta_k \leq \sum_{k \in K^0} \delta_k + \frac{2}{1 - \varepsilon_f} (\bar{f}(x_0) - \bar{f}^* + \Delta_{\max}),$$

where the last inequality follows from Lemma 4. Rearranging this completes the proof. $\square$

4.5 Bound for the case $\mu_k > 0$

Next, we show that the sum of δ_k over $k \in K^+$ is lower bounded. Recall that K^+ is the set of iteration indices k such that $\mu_k > 0$ as defined in eq. (5), and that μ_k is defined as $\theta_k \sqrt{\varsigma + \sum_{j \in K^+, j \leq k} \|g_j\|^2}$ with $\theta_k \in [\theta_{\min}, \theta_{\max}]$ in line 8 of Algorithm 1.

Before proceeding to the main proof, we present Lemma 6, which collects standard inequalities from the AdaGrad-Norm analysis of Ward et al. [46, Lemma 3.2] and Gratton et al. [19, Lemma 3.1]. Its proof is deferred to appendix A.

Lemma 6 *In Algorithm 1, we have*

$$\begin{aligned} \sum_{k \in K^+} \frac{\|g_k\|^2}{\mu_k^2} &\leq \frac{1}{\theta_{\min}^2} \left(\ln \left(\varsigma + \sum_{k \in K^+} \|g_k\|^2 \right) - \ln(\varsigma) \right), \\ \sum_{k \in K^+} \frac{\|g_k\|^2}{\mu_k} &\geq \frac{1}{\theta_{\max}} \left(\sqrt{\varsigma + \sum_{k \in K^+} \|g_k\|^2} - \sqrt{\varsigma} \right). \end{aligned}$$

We now lower bound the sum of δ_k over $k \in K^+$ with $\|g_k\|^2$. We define the following constants for later use:

$$C_1 := \frac{\bar{\alpha}}{\theta_{\max}}, \quad C_2 := \frac{M\bar{\alpha} + L/2}{\theta_{\min}^2}.$$

Lemma 7 *Under Assumptions 3 and 4, the sum of δ_k over $k \in K^+$ satisfies*

$$\sum_{k \in K^+} \delta_k \geq C_1 \left(\sqrt{\varsigma + \sum_{k \in K^+} \|g_k\|^2} - \sqrt{\varsigma} \right) - C_2 \left(\ln \left(\varsigma + \sum_{k \in K^+} \|g_k\|^2 \right) - \ln(\varsigma) \right).$$

Proof From the L -smoothness (Assumption 3), for $k \in K^+$ we have

$$\begin{aligned} \delta_k &= f(x_k) - f(x_k + \alpha_k d_k) && \text{(eq. (12) and } x_{k+1} = x_k + \alpha_k d_k) \\ &\geq -\alpha_k g_k^\top d_k - \frac{\alpha_k^2 L}{2} \|d_k\|^2 && \text{(Assumption 3)} \\ &\geq \left(\frac{\alpha_k}{M + \mu_k} - \frac{\alpha_k^2 L}{2(m + \mu_k)^2} \right) \|g_k\|^2 && \text{(eqs. (10) and (11))} \\ &\geq \left(\frac{\bar{\alpha}}{M + \mu_k} - \frac{L}{2(m + \mu_k)^2} \right) \|g_k\|^2. && \text{(Proposition 1 and } \alpha_k \leq 1) \end{aligned} \quad (24)$$

We can further bound the terms in eq. (24) as

$$\frac{\bar{\alpha}}{M + \mu_k} = \frac{\bar{\alpha}}{\mu_k} \frac{1}{1 + \frac{M}{\mu_k}} \geq \frac{\bar{\alpha}}{\mu_k} \left(1 - \frac{M}{\mu_k} \right) = \frac{\bar{\alpha}}{\mu_k} - \frac{M\bar{\alpha}}{\mu_k^2}, \quad \frac{L}{2(m + \mu_k)^2} \leq \frac{L}{2\mu_k^2}. \quad (25)$$

Thus, applying eq. (25) to eq. (24) yields

$$\delta_k \geq \left(\frac{\bar{\alpha}}{\mu_k} - \frac{M\bar{\alpha} + L/2}{\mu_k^2} \right) \|g_k\|^2. \quad (26)$$

Summing eq. (26) over $k \in K^+$ and applying Lemma 6 yield

$$\begin{aligned} \sum_{k \in K^+} \delta_k &\geq \sum_{k \in K^+} \left(\frac{\bar{\alpha}}{\mu_k} - \frac{M\bar{\alpha} + L/2}{\mu_k^2} \right) \|g_k\|^2 \\ &\geq \frac{\bar{\alpha}}{\theta_{\max}} \left(\sqrt{\varsigma + \sum_{k \in K^+} \|g_k\|^2} - \sqrt{\varsigma} \right) - \frac{M\bar{\alpha} + L/2}{\theta_{\min}^2} \left(\ln \left(\varsigma + \sum_{k \in K^+} \|g_k\|^2 \right) - \ln(\varsigma) \right), \end{aligned}$$

which yields the desired result. $\square$

4.6 Global Convergence Rate

Finally, we combine the results for $k \in K^0$ stated in Lemma 5 and $k \in K^+$ stated in Lemma 7 to derive an upper bound on the total sum of $\|g_k\|^2$. We define a constant F as follows:

$$F := f(x_0) - f^* + \frac{2}{1 - \varepsilon_f} \left(\bar{f}(x_0) - \bar{f}^* + \Delta_{\max} \right) + C_1 \sqrt{\varsigma} - C_2 \ln(\varsigma). \quad (27)$$

Then, we can obtain the following unified bound for the two types of iterations.

Lemma 8 *Under Assumptions 2 to 4, the sum of $\|g_k\|^2$ over $k \in K = K^0 \cup K^+$ satisfies*

$$F \geq \frac{c\bar{\alpha}}{M} \sum_{k \in K^0} \|g_k\|^2 + C_1 \sqrt{\varsigma + \sum_{k \in K^+} \|g_k\|^2} - C_2 \ln \left(\varsigma + \sum_{k \in K^+} \|g_k\|^2 \right),$$

Proof Since $f(x_{k_{\max}}) \geq f^*$ by Assumption 2 and $K = K^0 \cup K^+$, we have

$$f(x_0) - f^* \geq f(x_0) - f(x_{k_{\max}}) = \sum_{k \in K} \delta_k = \sum_{k \in K^0} \delta_k + \sum_{k \in K^+} \delta_k.$$

From Lemmas 5 and 7, we can further bound as

$$\begin{aligned} f(x_0) - f^* &\geq \frac{c\bar{\alpha}}{M} \sum_{k \in K^0} \|g_k\|^2 - \frac{2}{1 - \varepsilon_f} \left(\bar{f}(x_0) - \bar{f}^* + \Delta_{\max} \right) \\ &\quad + C_1 \sqrt{\varsigma + \sum_{k \in K^+} \|g_k\|^2} - C_1 \sqrt{\varsigma} - C_2 \ln \left(\varsigma + \sum_{k \in K^+} \|g_k\|^2 \right) + C_2 \ln(\varsigma). \end{aligned}$$

Substituting eq. (27) yields the desired bound, concluding the proof. $\square$

Now, we establish the following convergence-rate and iteration-complexity result for Algorithm 1.

Theorem 1 *Under Assumptions 1 to 4, the iterates generated by Algorithm 1 satisfy*

$$\frac{1}{k_{\max}} \sum_{k=0}^{k_{\max}-1} \|g_k\|^2 = \mathcal{O} \left(\frac{1}{k_{\max}} \right).$$

Consequently, for every $0 < \varepsilon \leq 1$, the algorithm generates an iterate satisfying $\|g_k\| \leq \varepsilon$ within $\mathcal{O}(1/\varepsilon^2)$ iterations.

Proof Let us define the function $\phi : \mathbb{R}_{>0} \rightarrow \mathbb{R}$ as

$$\phi(G) := \frac{C_1}{2} \sqrt{G} - C_2 \ln(G). \quad (28)$$

Since $\phi'(G) = \frac{C_1}{4\sqrt{G}} - \frac{C_2}{G}$, its minimum over $G > 0$ is attained at $G^* := \left(\frac{4C_2}{C_1}\right)^2$ with

$$\phi(G^*) = 2C_2 \left(1 - \ln\left(\frac{4C_2}{C_1}\right)\right) = 2C_2 \ln\left(\frac{eC_1}{4C_2}\right),$$

where e is the base of the natural logarithm. Thus, Lemma 8 yields

$$\begin{aligned} F &\geq \frac{c\bar{\alpha}}{M} \sum_{k \in K^0} \|g_k\|^2 + \frac{C_1}{2} \sqrt{\sum_{k \in K^+} \|g_k\|^2} + \phi\left(\sum_{k \in K^+} \|g_k\|^2 + \varsigma\right) \\ &\geq \frac{c\bar{\alpha}}{M} \sum_{k \in K^0} \|g_k\|^2 + \frac{C_1}{2} \sqrt{\sum_{k \in K^+} \|g_k\|^2} + \phi(G^*), \end{aligned}$$

which is equivalent to

$$0 \leq \frac{c\bar{\alpha}}{M} \sum_{k \in K^0} \|g_k\|^2 + \frac{C_1}{2} \sqrt{\sum_{k \in K^+} \|g_k\|^2} \leq F' \quad (29)$$

where

$$\begin{aligned} F' &:= F - \phi(G^*) \\ &= f(x_0) - f^* + \frac{2}{1 - \varepsilon_f} \left(\bar{f}(x_0) - \bar{f}^* + \Delta_{\max} \right) + C_1 \sqrt{\varsigma} - C_2 \ln(\varsigma) - 2C_2 \ln\left(\frac{eC_1}{4C_2}\right). \end{aligned}$$

It remains to bound $\sum_{k \in K} \|g_k\|^2 = \sum_{k \in K^0} \|g_k\|^2 + \sum_{k \in K^+} \|g_k\|^2$ under eq. (29). More generally, let $a, b > 0$, $c \geq 0$, and $x, y \geq 0$, and consider maximizing $x + y$ subject to $ax + b\sqrt{y} \leq c$. The maximum is attained on the boundary $ax + b\sqrt{y} = c$ because the objective is increasing in both variables. On this boundary, $y = (c - ax)^2/b^2$, so the objective is convex over $0 \leq x \leq c/a$ and attains its maximum at an endpoint. Evaluating the two endpoints gives $x + y \leq \max\{c/a, c^2/b^2\}$. Applying this bound to eq. (29) yields

$$\frac{1}{k_{\max}} \sum_{k=0}^{k_{\max}-1} \|g_k\|^2 \leq \frac{1}{k_{\max}} \max \left\{ \frac{MF'}{c\bar{\alpha}}, \frac{4(F')^2}{C_1^2} \right\}. \quad (30)$$

Since the minimum is no larger than the average, eq. (30) also guarantees an iterate satisfying $\|g_k\| \leq \varepsilon$ whenever

$$k_{\max} \geq \frac{1}{\varepsilon^2} \max \left\{ \frac{MF'}{c\bar{\alpha}}, \frac{4(F')^2}{C_1^2} \right\}. \quad (31)$$

This completes the proof. $\square$

As a remark, the function $\phi(G)$ in eq. (28) is closely related to the Lambert W function [10, 19]. For $x \in [-1/e, 0)$, the equation $ye^y = x$ admits two real negative solutions, and the smaller one is given by the second branch of the Lambert W function, $y = W_{-1}(x) < 0$. Using this function, the equation $\phi(G) = 0$ can be solved in closed form, which leads to a sharper bound in Theorem 1. Since this is not essential for the main argument, we omit the details and refer to [19].

We next interpret eq. (31) in the proof of Theorem 1. The contribution of the indices in K^0 is analogous to the classical analysis of gradient descent. Indeed, for gradient descent with fixed step size $\alpha = 1/L$, one has

$$\frac{1}{k_{\max}} \sum_{k=0}^{k_{\max}-1} \|g_k\|^2 \leq \frac{1}{k_{\max}} \frac{2(f(x_0) - f^*)}{\alpha}.$$

Thus, the term associated with K^0 reflects the standard $\mathcal{O}(f(x_0) - f^*)$ dependence of first-order methods since F' contains the initial gaps $f(x_0) - f^*$ and $\bar{f}(x_0) - \bar{f}^*$. In contrast, the contribution of K^+ resembles the behavior of AdaGrad-Norm, whose convergence bound scales quadratically with the initial optimality gap [46].

Finally, Theorem 1 implies the iteration-complexity bound

$$\min \{k \mid \|\nabla f(x_k)\| \leq \varepsilon_{\text{gtol}}\} = \mathcal{O}(1/\varepsilon_{\text{gtol}}^2).$$

The order $\mathcal{O}(1/\varepsilon_{\text{gtol}}^2)$ is worst-case optimal for first-order stationarity on the class of L -smooth nonconvex functions even when function values are exact [6]. Hence, bounded nonvanishing errors in the function values used by our hybrid mechanism do not worsen the best possible dependence on $\varepsilon_{\text{gtol}}$, although they affect the constants below. More explicitly, eq. (31) gives the sufficient bound

$$k_{\max} \geq \frac{1}{\varepsilon_{\text{gtol}}^2} \max \left\{ \frac{MF'}{c\bar{\alpha}}, \frac{4(F')^2}{C_1^2} \right\}.$$

Here $\bar{\alpha}$ depends only on c, L, m and the backtracking bounds, $C_1 = \bar{\alpha}/\theta_{\max}$, $C_2 = (M\bar{\alpha} + L/2)/\theta_{\min}^2$, and F' is displayed in the proof and contains ε_f , ς , the initial true and observed optimality gaps, and $\Delta_{\max}$. The construction in section 5.1 gives $m = \omega_{\min}$ and $M = (1 + (1 + p)\Lambda)\rho_0$, so the memory length and fixed initial scale enter the sufficient iteration bound through M . Thus the order symbol suppresses no dependence on the iteration counter, but it does suppress these problem and algorithm parameters. The guarantee deteriorates as the permitted function error approaches 1 through the factor $(1 - \varepsilon_f)^{-1}$.

5 Practical Choices of Algorithmic Variables

In sections 3 and 4, we introduced Algorithm 1 and established its convergence properties. In this section, we discuss practical choices of algorithmic variables and parameters that lead to fast and stable performance. The quantities to be discussed here are the matrices B_k , the step sizes α_k , and the regularization parameters μ_k .

5.1 Approximate Hessian B_k

We first discuss the construction of the matrices B_k used in line 9 of Algorithm 1. The purpose of this subsection is to show that a shifted raw L-BFGS construction with modified and carefully selected curvature pairs can satisfy Assumption 4.

Construction of B_k : We recall the L-BFGS construction [4, 28, 34]. For an iteration k , we define

$$s_k := x_{k+1} - x_k, \quad y_k := g_{k+1} - g_k$$

where g_k and g_{k+1} are the gradients at x_k and x_{k+1} , respectively. Starting from an initial matrix, an approximate Hessian is obtained by successively applying BFGS updates to a sequence of curvature pairs. Let $\{(s_{k_i}, y_{k_i})\}_{i=0}^{p-1}$ be a collection of curvature pairs, where k_i denotes the iteration index at which the i -th pair was generated. Starting from the initial matrix $B^{(0)} = \gamma I$ with $\gamma = \|y_{k_0}\|^2 / y_{k_0}^\top s_{k_0}$, we define BFGS($\{(s_{k_i}, y_{k_i})\}_{i=0}^{p-1}$) as the matrix $B^{(p)}$ obtained by sequentially applying the BFGS update as

$$B^{(j+1)} = B^{(j)} - \frac{B^{(j)} s_{k_j} s_{k_j}^\top B^{(j)}}{s_{k_j}^\top B^{(j)} s_{k_j}} + \frac{y_{k_j} y_{k_j}^\top}{y_{k_j}^\top s_{k_j}}$$

for $j = 0, 1, \dots, p-1$, where each curvature pair (s_{k_j}, y_{k_j}) is applied in order. Here, $p > 0$ denotes the maximum number of stored curvature pairs and is a fixed small integer in the L-BFGS method. When fewer than p curvature pairs are available, the BFGS update is applied using all currently available pairs.

We now provide a sufficient condition for Assumption 4 in the following proposition, which is a variant of existing results [15, Theorem 5.5] and [17, Lemma 1]. Its proof is deferred to appendix B.

Proposition 2 *Let $\{(s_{k_i}, y_{k_i})\}_{i=0}^{p-1}$ be a fixed number of curvature pairs with $s_{k_i} \neq 0$. Assume that there exists a constant $\Lambda > 0$ such that every pair satisfies*

$$y^\top s \geq \frac{1}{\Lambda} \|y\|^2 > 0. \quad (32)$$

Then

$$0 \prec \text{BFGS}(\{s_{k_i}, y_{k_i}\}_{i=0}^{p-1}) \preceq (1+p)\Lambda I. \quad (33)$$

Proposition 2 gives the upper spectral bound needed below. A fixed diagonal shift provides the lower bound in Assumption 4.

We select curvature pairs using a cautious update in the spirit of curvature-based cautious updates [26, 29, 30]. Define the fixed initial scale and diagonal shift by

$$\rho_0 := \max\{1, \|g_0\|\}, \quad \omega_{\min} := \min\{1, \sqrt{\zeta}\} > 0.$$

After damping, a raw pair $(s, \bar{y})$ is retained only if

$$\bar{y}^\top s \geq \max\left\{\frac{\|\bar{y}\|^2}{10^8 \rho_0}, 10^{-16}\right\}. \quad (34)$$

For the proof only, define $\widehat{y} := \bar{y}/\rho_0$. Then every stored pair $(s, \widehat{y})$ satisfies eq. (32) with the iteration-independent constant $\Lambda = 10^8$. The absolute floor rejects numerically degenerate pairs but is not needed for the upper bound. The homogeneity of the BFGS update gives

$$\text{BFGS}(\{(s_{k_i}, \bar{y}_{k_i})\}) = \rho_0 \text{BFGS}(\{(s_{k_i}, \widehat{y}_{k_i})\}).$$

Using at most $p = 10$ pairs, we set

$$B_k = \begin{cases} \omega_{\min} I + \text{BFGS}(\{(s_{k_i}, \bar{y}_{k_i})\}), & \text{if the memory is nonempty,} \\ (\omega_{\min} + \rho_0) I, & \text{otherwise.} \end{cases} \quad (35)$$

By Proposition 2, the homogeneity identity, and $\omega_{\min} \leq 1 \leq \rho_0$, it follows that

$$\omega_{\min} I \preceq B_k \preceq (1 + (1 + p)\Lambda)\rho_0 I.$$

Thus Assumption 4 holds with the iteration-independent constants $m = \omega_{\min}$ and $M = (1 + (1 + p)\Lambda)\rho_0$. For a candidate pair with $y_{k_i}^\top s_{k_i} < 0$, the implementation applies a scalar Powell-type damping before the cautious test. Specifically, it sets $b_i = \|y_{k_i}\|^2 / |y_{k_i}^\top s_{k_i}|$ and

$$\bar{y}_{k_i} = \theta_i^{\text{damp}} y_{k_i} + (1 - \theta_i^{\text{damp}}) b_i s_{k_i}, \quad \theta_i^{\text{damp}} = \frac{0.8 b_i \|s_{k_i}\|^2}{b_i \|s_{k_i}\|^2 - y_{k_i}^\top s_{k_i}} \in [0, 1).$$

If $y_{k_i}^\top s_{k_i} \geq 0$, it sets $\bar{y}_{k_i} = y_{k_i}$. Candidates for which this scalar damping is numerically unsafe are discarded before eq. (34). This is the damping rule used in the released code. The fixed initial-scale upper test and the diagonal shift, rather than damping itself, guarantee the two bounds in Assumption 4. No convexity or cocoercivity assumption on ∇f is used.

Remarks for Practical Implementation: Having established that the proposed construction of B_k satisfies Assumption 4, we now present several remarks on its practical implementation. We begin by describing how to compute the search direction

$$d_k = -(B_k + \mu_k I)^{-1} g_k$$

with B_k in eq. (35). We use a compact representation of the raw L-BFGS matrix to solve the regularized system exactly [25]. In particular, the implementation computes $d_k = -(B_k + \mu_k I)^{-1} g_k$ for the matrix B_k defined in eq. (35), without replacing the additive diagonal regularization by modified curvature pairs. We adopted this exact compact solve to align the implemented direction directly with the direction analyzed in section 4. When the memory is empty, the same definition reduces to $d_k = -g_k/(\rho_0 + \omega_{\min} + \mu_k)$. If the compact factorization fails numerically or does not produce a finite descent direction, the implementation uses this zero-memory direction, which also corresponds to a matrix satisfying Assumption 4. For comparison, the implementation retains the simpler shifted-pair approximation

$$B_k + \mu_k I \approx \text{BFGS}(\{s_{k_i}, \bar{y}_{k_i} + (\mu_k + \omega_{\min}) s_{k_i}\}_{i=0}^{p-1}). \quad (36)$$

The variants using this approximation are denoted by the suffix -SP and are treated as heuristic variants rather than as the implementations covered directly by the analysis. They retain the standard two-loop recursion and are included as lower-overhead practical alternatives to the exact compact solve.

Separately, there exist function-based modified secant conditions [1, 22, 48, 51] that adjust the secant equation using available function values when those quantities are reliable. Such adjustments aim to improve practical convergence behavior. We partially adopt these function-based modifications following prior work [49, 52]. The suffix -MS in section 6 indicates the use of this technique. Because these modifications are well established, we omit low-level implementation details here. For the proposed -MS variants, a scale-aware safeguard sets the function-based correction σ_i to zero whenever $|\sigma_i| > \sqrt{\|y_{k_i}\|^2 / \|s_{k_i}\|^2}$, preventing the correction from dominating the raw gradient difference. Further specifics are available in the cited references and in our open-source implementation.

5.2 Step Size α_k

We next discuss the adjustment of the step size α_k in line 4 of Algorithm 2. As indicated in the pseudo-code, we choose α_k using interpolation of the objective function. Moré–Thuente line searches [31] are widely used in quasi-Newton methods [44], and they typically employ cubic or quadratic interpolation to select trial step sizes. In our implementation, we use an essentially identical procedure, except that the trial step size is obtained by clipping it to the interval $[\beta_{\min} \alpha_k, \beta_{\max} \alpha_k]$. For all numerical experiments, we set the Armijo parameter to $c = 10^{-4}$ and choose the interpolation bounds as $\beta_{\min} = 1/16$ and $\beta_{\max} = 15/16$.

Furthermore, when $\mu_k > 0$ and $\alpha_k = 1$, we introduce a one-time adjustment of the trial step size using gradient information. Let d_k be the search direction and g_k and g_{try} the gradients at the current and trial points. If the directional derivative along d_k changes sign, $d_k^\top g_k < 0$ and $d_k^\top g_{\text{try}} > 0$, and the trial gradient is sufficiently aligned with d_k , namely $d_k^\top g_{\text{try}} > 0.5 \|d_k\| \|g_{\text{try}}\|$, we rescale α_k once by a secant-like factor,

$$\alpha_k \leftarrow \text{clip} \left(\alpha_k \frac{-d_k^\top g_k}{d_k^\top g_{\text{try}} - d_k^\top g_k}, \beta_{\min} \alpha_k, \beta_{\max} \alpha_k \right),$$

where $\text{clip}(x, a, b) := \min(\max(x, a), b)$ is a clipping function. When objective values are unreliable and backtracking and interpolation do not occur, this correction effectively refines the step size. Since this adjustment is at most once per iteration, the theoretical results in section 4 remain valid.

5.3 Regularization Parameter μ_k

We finally discuss the adjustment of the regularization parameter μ_k in line 8 of Algorithm 1. In our implementation, the OFFO accumulator contains exactly those gra-

dient norms generated during regularized iterations:

$$G_k^2 = \varsigma + \sum_{j \in K^+, j \leq k} \|g_j\|^2, \quad \mu_k = \text{clip} \left(\frac{1}{10} \|g_k\|, \frac{1}{100} G_k, G_k \right).$$

Hence $\mu_k = \theta_k G_k$ with $\theta_k \in [10^{-2}, 1]$, exactly as required by eq. (7). We set $\varsigma = 10^{-20}$, so that its square root, the initial scale G_{-1} , equals 10^{-10} . The primary implementation sets the permitted number of restarts to zero and therefore coincides exactly with the analyzed accumulator. The restart was originally intended only as a practical heuristic rather than as an essential part of the proposed algorithm. The heuristic is motivated by the observation that continued growth of G_k can over-regularize later steps after the method has again obtained clear progress in the objective value. Resetting the accumulator after such progress is intended to reduce the effective regularization and recover faster quasi-Newton behavior. For the optional heuristic, a fixed threshold $\tau > 0$ resets only the accumulator in eq. (7) when the method returns from K^+ to K^0 and

$$\min_{j \in K^0, j < k} (\bar{f}(x_j) - \Delta_j) - \bar{f}(x_k) \geq \tau. \quad (37)$$

The sets K^0 and K^+ and the reference values in eq. (37) are not reset. Because this optional rule is heuristic, all primary experiments use the analyzed no-restart method. The variant with at most ten restarts is reserved for the separate appendix ablation in appendix C.

6 Numerical Experiments

We conducted numerical experiments to evaluate the practical performance of our proposed method and to compare it with existing optimization algorithms.

6.1 Methods

We compared the following methods in our experiments.

- (Ours): Our proposed regularized quasi-Newton method
 - We used Algorithms 1 and 2 with the details in section 5.
 - (Ours-MS) means the variant incorporating the modified secant condition described in section 5.1.
 - The suffix -SP denotes the corresponding heuristic variant that replaces the exact compact solve by the shifted-pair approximation in eq. (36).
- (Line): Line search L-BFGS method
 - Our Python implementation ported from LBFGSpp [35].
 - The step size was determined using the Moré–Thuente line search, closely resembling SciPy’s implementation.
 - (Line-MS) means the same variant as Ours-MS.
- (Reg): Regularized L-BFGS method [25]

- A quadratic regularization-based quasi-Newton method that does not rely on line search and is conceptually closer to trust-region methods.
- We wrapped the function `regLBFGS` with `solveNonmonotone` from [25].
- (Reg-Sec) denotes the variant using the approximate computation in eq. (36). It wraps the `regLBFGSsec` function from [25].
- (SciPy): SciPy’s L-BFGS-B method [44]
 - An established L-BFGS method implemented in C and called from Python.
 - We used the default parameters, with the exception that `ftol` was set to 0 to avoid premature termination.
- (NTQN): Noise-tolerant quasi-Newton method [40, 47]
 - A method designed to accommodate inexact gradient information, as mentioned in section 2.3.
- (ASTR1-Adagrad): The purely first-order adagrad variant of ASTR1 reported by Gratton et al. [19, Table 1], using $B_k = 0$, $\gamma_k = 1$, $\mu = 1/2$, $\varsigma = 1/100$, and $\vartheta = \theta = 1$. Its steps and stopping decision use only gradients. The shared callback records the initial objective value solely for diagnostics and does not expose it to the method.

For the primary NTQN comparison, we use the wrapper’s common gradient-tolerance stopping rule. We additionally run the authors’ recommended termination mode (`terminate=3`) with that wrapper-level stop disabled. In the function-noise experiment, NTQN receives the conservative bound $\text{eps}_f = 10^{-2}$ for perturbations bounded by 10^{-3} , matching the value of ε_f supplied to the proposed method. NTQN uses this bound in both its noise-tolerant line search and its recommended termination test. In the recommended mode, termination is controlled by NTQN itself, while the common gradient tolerance is applied only afterward to its recorded trajectory for uniform scoring. Because NTQN may appropriately stop at its estimated noise neighborhood before attaining that tolerance, we report both solver-declared termination and attainment of the common scoring tolerance and distinguish such early termination from poor oracle efficiency.

In the following, each method is referred to by the name given in parentheses. Except for the first-order ASTR1-Adagrad baseline, all these methods are L-BFGS-based, and we set the number of stored curvature pairs to 10, the default value in SciPy’s L-BFGS-B implementation. All experiments were conducted on a Microsoft Windows 11 Home system (version 10.0, build 26100) equipped with a 13th Gen Intel® Core™ i7-1360P CPU, featuring 12 physical cores and 16 logical processors. All computations were performed using the CPU only.

6.2 Experimental Results

6.2.1 Test Problems and Settings

We evaluated all algorithms on unconstrained problems from the CUTEst benchmark collection [16], implemented in Python using the PyCUTEst interface [14]. All problems were initialized with the default starting points provided by the library. Following previous work [25], we excluded problems whose initial points were already

stationary or whose function values or gradients contained NaN or INF. The 64- and 32-bit collections contain 220 problems with dimensions from 2 to 123,200, and the 16-bit collection contains 152 problems over the same dimension range. The supplied error rates are 2.22×10^{-9} , 1.19×10^{-3} , and 9.77×10^{-2} for the 64-, 32-, and 16-bit experiments, respectively. The 64-bit value equals SciPy’s default heuristic $10^7 \epsilon_{\text{mach}}$. At lower precisions we reduce the multiplier to 10^4 and 10^2 , respectively, because retaining 10^7 would violate the required range $\epsilon_f < 1$ and yield an uninformative error allowance. These multipliers are practical safety margins rather than estimates supplied by the theory, which motivates the misspecification study below. In addition to these precision experiments, we distinguish two additive-noise experiments on the 64-bit collection:

1. a theory-aligned setting with $\bar{f}(x) = f(x) + \xi_f$, $\xi_f \sim \text{unif}(-10^{-3}, 10^{-3})$, and exact gradients, and
2. an adverse stress test with the same function perturbation and independent componentwise gradient perturbations $\xi_g^{(i)} \sim \text{unif}(-10^{-3}, 10^{-3})$.

We supply $\epsilon_f = 10^{-2}$ in both settings and additionally test 10^{-4} and 10^{-1} to assess under- and overestimation of this error parameter. Underestimating ϵ_f makes Δ_k too small to absorb the actual function error, makes the K^0 test easier to satisfy, and can therefore cause the method to trust unreliable comparisons and take more unregularized steps outside the theorem’s error model. Overestimating ϵ_f preserves the error model and hence is safe from the viewpoint of the convergence theory, provided that $\epsilon_f < 1$. It relaxes Armijo acceptance through a larger Δ_k , but it also lowers every reference value $\bar{f}(x_j) - \Delta_j$ and thereby makes subsequent entry into K^0 harder. The latter effect can keep the method in K^+ , where the nondecreasing scale G_k produces more conservative OFFO-type steps. Thus overestimation is expected to preserve stability but can reduce practical efficiency, whereas underestimation can place the computation outside the assumed error model. The sensitivity experiment examines this distinction. Each noisy experiment uses the five seeds 0, 1, 2, 3, 4, records the seed, and supplies the same seed to all solvers for a given problem and scenario. The joint-noise setting deliberately violates eq. (3). It is included to test performance under an unfavorable perturbation, not to validate the exact-gradient theorem. This setting is particularly adverse for our method because its theory and regularization logic rely on accurate gradient information, whereas NTQN was designed specifically to accommodate bounded gradient errors.

Each setting uses a different error rate ϵ_f in the error model (eq. (2)), as summarized in Table 2. The table relates the machine epsilon (maximum relative rounding error) for each floating-point precision to the ϵ_f value used in our experiments. The relatively large ϵ_f values are chosen to account for the cumulative errors that may arise during optimization. We first run SciPy with a limit of 15,000 iterations on each problem. If it attains the scenario’s gradient tolerance after k_{SciPy} iterations, every other method is limited to $\min\{15,000, 15k_{\text{SciPy}}\}$ iterations. Otherwise, the limit is 15,000. We timed out runs exceeding 10 minutes per problem.

explanation	min abs value	relative error	ε_f
64-bit double precision	2.23×10^{-308}	$2^{-52} \approx 2.22 \times 10^{-16}$	2.22×10^{-9}
32-bit single precision	1.18×10^{-38}	$2^{-23} \approx 1.19 \times 10^{-7}$	1.19×10^{-3}
16-bit half precision	6.10×10^{-5}	$2^{-10} \approx 9.77 \times 10^{-4}$	9.77×10^{-2}

Table 2: The “min abs value” is the smallest positive normalized number, the “relative error” is the maximum relative rounding error, and ε_f is the parameter in (2) used in our experiments.

6.2.2 Performance Profiles

We used performance profiles [12] to compare the algorithms. Let P denote the set of test problems and S the set of solvers. For each problem $p \in P$ and solver $s \in S$, let $n_{p,s} > 0$ denote the number of oracle calls (i.e., function and gradient evaluations) required to reach a specified gradient norm tolerance $\varepsilon_{\text{gtol}}$. In this experiment, we use the number of oracle calls rather than clock time to avoid the influence of algorithmic overhead and to focus on the intrinsic optimization efficiency. When a solver fails to converge within the gradient tolerance or exceeds a maximum number of oracle calls, we set $n_{p,s} = +\infty$. Then, the performance ratio $r_{p,s}$ is defined as

$$r_{p,s} = \frac{n_{p,s}}{\min_{s' \in S} n_{p,s'}}.$$

For a given threshold $\tau \geq 1$ and solver s , the performance profile plots the proportion of problems for which $r_{p,s} \leq \tau$. Thus, the y -coordinate for solver s at x -coordinate τ is given by

$$\rho_s(\tau) = \frac{1}{|P|} |\{p \in P \mid r_{p,s} \leq \tau\}|.$$

A curve that lies above others indicates better performance. Following standard implementations [44], we used the infinity norm for the gradient, though other norms may equally be employed. All reported success percentages use the same common gradient tolerance as the corresponding profiles. A run that terminates without attaining that tolerance is counted as unsolved for this metric.

6.2.3 Results with Artificial Noise

We first separate the theory-aligned function-noise experiment from the joint-noise stress test described in section 6.2.1. The former tests the assumptions of section 4. The latter asks whether the implementation remains useful when both available quantities are perturbed. Therefore, success in the joint-noise experiment is evidence that the method can remain effective even in a setting that is theoretically unfavorable to it, but it is not presented as validation of the exact-gradient complexity result. The experiment notebook compares the hybrid method with the first-order ASTR1-Adagrad baseline of Gratton et al. [19] and varies the supplied ε_f over two orders of magnitude around its nominal value. All main-text profiles use the analyzed no-restart

method. The earlier accumulator-restart heuristic is evaluated separately in the appendix. Across the 220 problems and five paired seeds, Ours and Ours-MS attain $\varepsilon_{\text{gtol}} = 10^{-2}$ on 85.1% and 85.5%, respectively, of the 1,100 function-noise instances. Their seed-to-seed standard deviations in the solved fraction are only 0.60 and 0.18 percentage points. The corresponding rates are 81.2% for NTQN and 77.8% for Reg. They are 74.0% for Reg-Sec and 41.5% for ASTR1-Adagrad. Among successful runs, the median oracle-call counts are 62 for Ours, 67 for Ours-MS, 64 for NTQN, and 131 for ASTR1-Adagrad. The shifted-pair variant Ours-SP solves 83.9%, while the compact solver records no fallback for Ours and nine fallbacks in total for Ours-MS across these instances. The complete distributions, including failures, are represented by the profiles in Figure 1. Both proposed variants solve at least one instance of dimension 123,200, the largest dimension in the retained collection. In the joint-noise stress test, Ours and Ours-MS solve 83.6% and 84.8% of the instances, compared with 71.9% for NTQN, 73.5% for Reg, and 42.8% for ASTR1-Adagrad. The two proposed rates have seed-to-seed standard deviations of 0.64 and 0.22 percentage points. The corresponding Ours-SP rate is 80.6%, and the compact solver records two and one fallbacks for Ours and Ours-MS, respectively. Here the componentwise gradient perturbation 10^{-3} is one order below the common scoring tolerance, and the regularized K^+ steps damp the effect of an inaccurate gradient when function comparisons cease to certify progress. This offers a plausible explanation for the strong empirical result, including the higher solved fraction than NTQN under the common stopping criterion, but it does not establish gradient-noise tolerance or extend the exact-gradient theorem. Under NTQN’s recommended termination mode, 50.1% of the same function-noise instances attain the common tolerance, compared with 81.2% under the wrapper’s common-tolerance stop. All recommended-mode runs terminate without timeout or exception, and 619 of them return flag 4. The lower scored fraction therefore mainly reflects NTQN’s noise-dependent stopping rule rather than failed execution.

For the error-bound sensitivity study, underestimating ε_f as 10^{-4} gives solved fractions of 87.1% for Ours and 81.9% for Ours-MS, whereas the nominal 10^{-2} gives 85.1% and 85.5%, and overestimating it as 10^{-1} gives 75.5% and 74.7%. Thus, underestimation happens to improve the aggregate success rate of the unmodified variant but harms the modified-secant variant, while the tenfold overestimate reduces the rates by 9.6 and 10.8 percentage points, respectively, relative to the nominal setting. The relatively high solved fractions remain plausible because this experiment uses exact gradients, its termination tolerance $\varepsilon_{\text{gtol}} = 10^{-2}$ exceeds the function-noise level by one order of magnitude, and function-value comparisons remain informative when the actual decrease dominates the noise. For $\varepsilon_f = 10^{-1}$, the multiplier $2\varepsilon_f/(1 - \varepsilon_f)$ in Δ_k equals approximately 0.222, more than one hundred times the actual additive-noise level. The resulting reference shift keeps the method in K^+ more often and produces conservative steps through the accumulated regularization scale. Although this behavior does not invalidate the theory, the results show that a safety margin that is much larger than the actual error can reduce practical efficiency. By contrast, underestimation can violate the assumed error bound even though it happens to improve the aggregate success rate of the unmodified variant in this experiment. This asymmetric response supports treating ε_f as a practically consequential error allowance

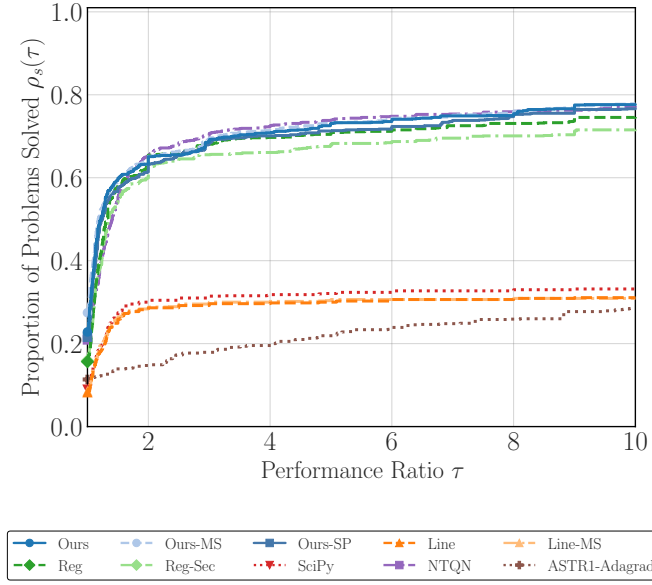


Fig. 1: Theory-aligned function-value-noise results at noise level 10^{-3} , supplied error bound $\varepsilon_f = 10^{-2}$, and $\varepsilon_{\text{gtol}} = 10^{-2}$.

rather than as a parameter for which either direction of misspecification is uniformly advantageous.

6.2.4 Results at Different Precision Levels

We next present performance profiles for each floating-point precision (64-, 32-, and 16-bit). We evaluated each method at three gradient norm tolerance levels, i.e., $\varepsilon_{\text{gtol}} \in \{10^{-1}, 10^{-3}, 10^{-5}\}$. These tolerance levels represent different optimization regimes, from moderate accuracy to high precision requirements. Figure 4 shows the comprehensive performance profiles across all three precision levels. At the strictest tolerance $\varepsilon_{\text{gtol}} = 10^{-5}$, Ours and Ours-MS solve 84.5% and 85.5% of the 64-bit problems, 52.3% and 52.7% of the 32-bit problems, and 28.9% and 28.9% of the 16-bit problems. The decreasing rates quantify the expected loss of attainable accuracy as arithmetic precision is reduced, while both variants remain among the most robust methods in the profiles. Their relative ranking depends on the tolerance and precision, so we avoid claiming uniform superiority and report both the fraction solved and the performance ratios in Figure 4.

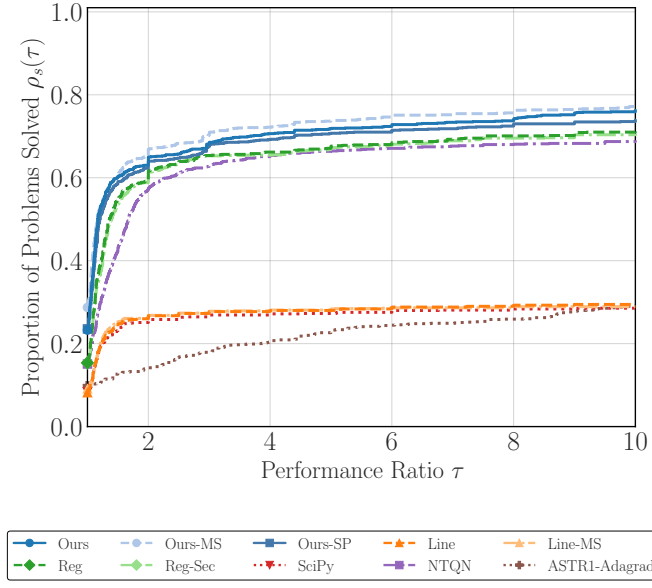


Fig. 2: Performance profile for the deliberately adverse joint function-and-gradient noise setting (componentwise level 10^{-3}) and $\epsilon_{\text{gtol}} = 10^{-2}$. This experiment lies outside the exact-gradient theory.

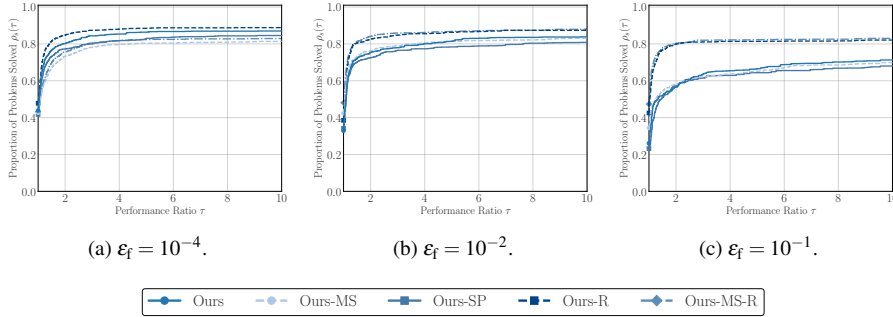


Fig. 3: Performance profiles for error-bound misspecification under function-value-only noise of level 10^{-3} and $\epsilon_{\text{gtol}} = 10^{-2}$.

6.3 Computational Time per Iteration

In this subsection, we present an experiment comparing the per-iteration computational cost of each method. Here, we measure total execution time rather than the number of oracle calls. We conducted experiments on the ill-conditioned quadratic

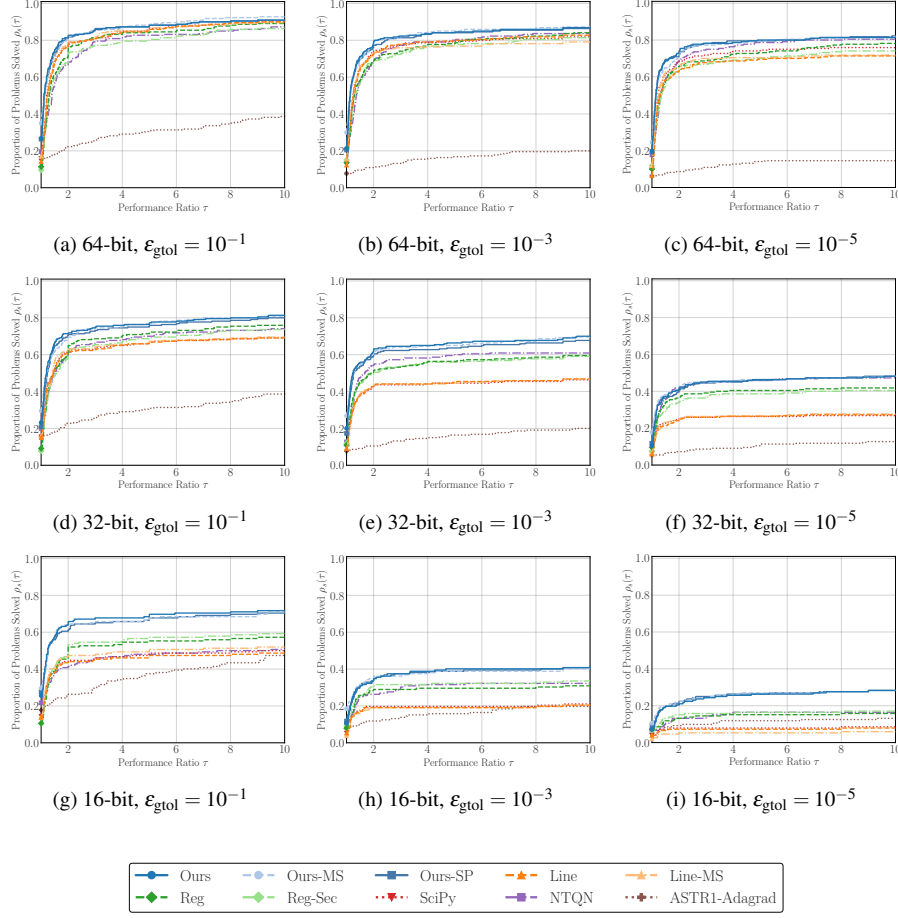


Fig. 4: Performance profiles across floating-point precisions (64-, 32-, and 16-bit) and tolerance levels.

problem

$$\underset{x \in \mathbb{R}^n}{\text{minimize}} \quad \frac{1}{2} \sum_{i=1}^n ix_i^2$$

with $n = 10,000$. The initial point x_0 was chosen as the all-ones vector. We used memory size $m = 10$ and maximum iterations $k_{\max} = 100$, and we did not set any stopping criteria other than reaching $k_{\max}$. The simplicity of the optimization problem and the small value of $k_{\max}$ are well suited to the purpose of this experiment, as they ensure that the total runtime is dominated by algorithmic overhead rather than function or gradient computation. Timing data were collected after three warm-up runs, followed by 100 recorded trials. BLAS operations were restricted to one thread to avoid method-dependent effects from dynamic thread scheduling.

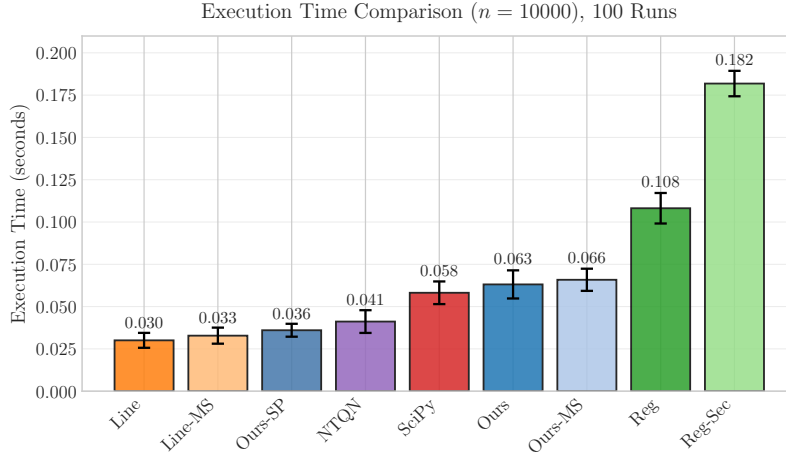


Fig. 5: Mean execution times for the experiment in section 6.3. This plot roughly indicates the extra computational overhead of each method. Error bars represent one standard deviation over 100 runs.

	Line	Line-MS	Ours-SP	NTQN	SciPy	Ours	Ours-MS	Reg	Reg-Sec
# Calls	212	212	202	219	212	202	202	206	206
$\bar{f}(x_{k_{\max}})$	1.24	1.24	1.34	1.43	1.24	1.34	1.34	1.47	1.48

Table 3: The number of oracle calls and final observed objective values for the experiment in section 6.3.

In Figure 5, we report the execution time statistics: the shifted-pair Ours-SP has per-iteration cost close to the least expensive Python implementations, whereas the exact compact solves in Ours and Ours-MS take about 1.8 times its runtime in this isolated test. This timing experiment measures algorithmic overhead under deliberately inexpensive objective and gradient evaluations and does not by itself establish end-to-end robustness or superiority in oracle cost. Because wall-clock time depends strongly on implementation details and the computing environment, we restrict its interpretation to this isolated experiment and do not draw broader conclusions from the timing results. For a fairer comparison of algorithmic efficiency, we place greater emphasis on oracle-call counts and success profiles.

In Table 3, we report the total number of oracle calls (calls of $\bar{f}$ and g) and final objective values $\bar{f}(x_{k_{\max}})$. The purpose of this table is to check whether the methods achieved similar convergence behavior within the limited number of iterations. Comparable values would indicate that differences in convergence behavior do not dominate the execution-time comparison.

7 Conclusion

We proposed a hybrid regularized quasi-Newton method for smooth nonconvex optimization with bounded errors in objective-function values. Under exact gradients and uniformly bounded positive definite quasi-Newton matrices, the method retains the order-optimal $\mathcal{O}(1/\varepsilon^2)$ first-order iteration complexity of the noise-free L -smooth nonconvex setting while selectively using noisy function values. The implementation enforces the fixed-initial-scale upper-curvature condition on raw pairs, supplies the lower bound through a fixed diagonal shift, and uses the same non-restarted OFFO accumulator as the analysis. The experiments distinguish function-value noise covered by the theorem from a joint-noise stress test outside its scope and assess sensitivity to the supplied error bound. An analysis with inexact gradients, which can at best guarantee convergence to a noise-dependent neighborhood without additional assumptions, remains future work.

A Proof of Lemma 6

The proof of Lemma 6 is similar to Lemma 3.2 of Ward et al. [46] and Lemma 3.1 of Gratton et al. [19].

Proof Recall that eq. (7) defines μ_k as $\theta_k \sqrt{\varsigma + \sum_{j \in K^+, j \leq k} \|g_j\|^2}$, and line 8 of Algorithm 1 asserts $\theta_k \in [\theta_{\min}, \theta_{\max}]$. Let $k_i \in K^+$ be the i -th element of K^+ . Define $S_{-1} := \varsigma$ and $S_i := \varsigma + \sum_{\ell=0}^i \|g_{k_\ell}\|^2$ for $i \geq 0$. For any $0 < x < y$, $x/y \leq -\ln(1 - x/y) = \ln y - \ln(y - x)$ holds. Thus, we have

$$\begin{aligned} \frac{\|g_{k_i}\|^2}{\mu_{k_i}^2} &\leq \frac{1}{\theta_{\min}^2} \frac{\|g_{k_i}\|^2}{S_i} && \text{(definition)} \\ &\leq \frac{1}{\theta_{\min}^2} (\ln S_i - \ln S_{i-1}). && (x/y \leq \ln y - \ln(y - x)) \end{aligned}$$

Similarly, $x/\sqrt{y} \geq x/(\sqrt{y} + \sqrt{y-x}) = \sqrt{y} - \sqrt{y-x}$ holds. Thus, we have

$$\begin{aligned} \frac{\|g_{k_i}\|^2}{\mu_{k_i}} &\geq \frac{1}{\theta_{\max}} \frac{\|g_{k_i}\|^2}{\sqrt{S_i}} && \text{(definition)} \\ &\geq \frac{1}{\theta_{\max}} (\sqrt{S_i} - \sqrt{S_{i-1}}). && (x/\sqrt{y} \geq \sqrt{y} - \sqrt{y-x}) \end{aligned}$$

Summing these inequalities over $0 \leq i < |K^+|$ yields the claim. $\square$

B Proof of Proposition 2

The proof of Proposition 2 uses the standard BFGS update.

Proof From eq. (32), the initial scalar satisfies $0 < \gamma = \|y_{k_0}\|^2 / (y_{k_0}^\top s_{k_0}) \leq \Lambda$. Let B and B_+ denote the positive definite Hessian approximations before and after one update. The BFGS update rule is

$$B_+ = B - \frac{B s s^\top B}{s^\top B s} + \frac{y y^\top}{y^\top s}.$$

Positive definiteness follows from $B \succ 0$, $s \neq 0$, and $y^\top s > 0$. Moreover, eq. (32) and the fact that $B s s^\top B / (s^\top B s)$ is positive semidefinite give

$$B_+ \preceq B + \frac{y y^\top}{y^\top s} \preceq B + \Lambda I.$$

Applying this inequality recursively over p updates from $B^{(0)} = \gamma I \preceq \Lambda I$ proves eq. (33). $\square$

The fixed-initial-scale upper-curvature inequality is enforced directly by eq. (34), while the fixed shift in eq. (35) supplies the lower spectral bound. No convexity or cocoercivity assumption is used. This point is important because the convergence theorem permits nonconvex objectives.

C Accumulator-Restart Ablation

This appendix isolates the earlier accumulator-restart heuristic from the primary no-restart experiments that match Algorithm 1 and the analysis in section 4. The restart variant applies the condition in eq. (37) only when returning from K^+ to K^0 , permits at most ten restarts, and otherwise uses the same problems, seeds, stopping rules, and parameter values as the corresponding no-restart method. For each setting, the comparison records oracle calls, failures, the distribution of the number of restarts, and whether the cap is reached. Over the 1,100 function-noise instances, the no-restart and restart variants attain $\varepsilon_{\text{tol}} = 10^{-2}$ on 85.1% and 87.6%, respectively, corresponding to 164 and 136 instances that do not attain the tolerance. Among the 928 instances solved by both variants, restart uses fewer oracle calls on 185, the same number on 654, and more on 89. The median call ratio is 1.00, and its geometric mean is 0.895. The restart count has mean 1.48, median zero, and maximum ten, with the cap reached on 62 instances. Hence the heuristic gives a small aggregate improvement but is inactive on at least half of the instances and reaches its imposed cap often enough that these results do not justify incorporating it into the theoretical algorithm.

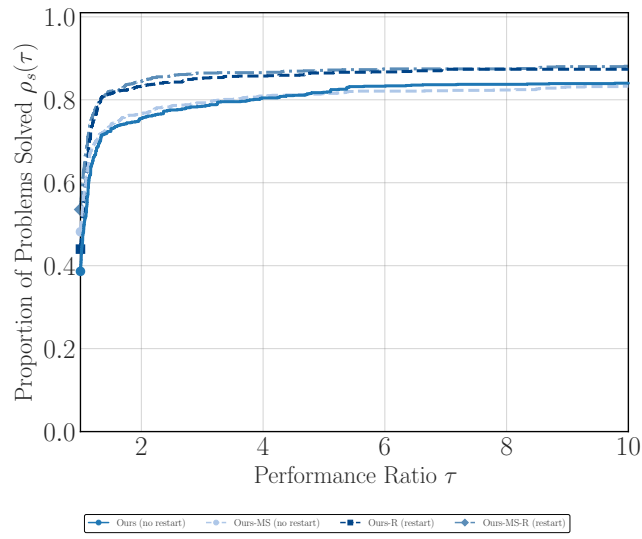


Fig. 6: Performance profile for the restart ablation on the same 1,100 paired function-noise instances as the primary experiment. Ours is the analyzed method with no accumulator restart. Ours-R applies the optional rule in eq. (37) with at most ten restarts. The close curves agree with the paired call counts: restart changes aggregate success only slightly and is not uniformly faster.

References

1. Babaie-Kafaki, S., Ghanbari, R., Mahdavi-Amiri, N.: Two new conjugate gradient methods based on modified secant equations. *Journal of Computational and Applied Mathematics* **234**(5), 1374–1386 (2010). DOI 10.1016/j.cam.2010.01.052
2. Bellavia, S., Gurioli, G., Morini, B., Toint, P.L.: The Impact of Noise on Evaluation Complexity: The Deterministic Trust-Region Case. *Journal of Optimization Theory and Applications* **196**(2), 700–729 (2023). DOI 10.1007/s10957-022-02153-5
3. Berahas, A.S., Cao, L., Choromanski, K., Scheinberg, K.: A Theoretical and Empirical Comparison of Gradient Approximations in Derivative-Free Optimization. *Foundations of Computational Mathematics* **22**(2), 507–560 (2022). DOI 10.1007/s10208-021-09513-z
4. Berahas, A.S., Curtis, F.E., Zhou, B.: Limited-memory BFGS with displacement aggregation. *Mathematical Programming* **194**(1), 121–157 (2022). DOI 10.1007/s10107-021-01621-6
5. Caflisch, R.E.: Monte Carlo and quasi-Monte Carlo methods. *Acta Numerica* **7**, 1–49 (1998). DOI 10.1017/S0962492900002804
6. Carmon, Y., Duchi, J.C., Hinder, O., Sidford, A.: Lower bounds for finding stationary points I. *Mathematical Programming* **184**(1), 71–120 (2020). DOI 10.1007/s10107-019-01406-y
7. Carter, R.G.: Numerical Experience with a Class of Algorithms for Nonlinear Optimization Using Inexact Function and Gradient Information. *SIAM Journal on Scientific Computing* **14**(2), 368–388 (1993). DOI 10.1137/0914023
8. Cartis, C., Sampaio, Ph.R., Toint, Ph.L.: Worst-case evaluation complexity of non-monotone gradient-related algorithms for unconstrained optimization. *Optimization* **64**(5), 1349–1361 (2015). DOI 10.1080/02331934.2013.869809
9. Clancy, R.J., Menickelly, M., Hückelheim, J., Hovland, P., Nalluri, P., Gjini, R.: TROPHY: Trust Region Optimization Using a Precision Hierarchy. In: *Computational Science – ICCS 2022*, pp. 445–459. Springer, Cham (2022). DOI 10.1007/978-3-031-08751-6_32
10. Corless, R.M., Gonnet, G.H., Hare, D.E.G., Jeffrey, D.J., Knuth, D.E.: On the Lambert W function. *Advances in Computational Mathematics* **5**(1), 329–359 (1996). DOI 10.1007/BF02124750
11. Doikov, N., Mishchenko, K., Nesterov, Y.: Super-Universal Regularized Newton Method. *SIAM Journal on Optimization* **34**(1), 27–56 (2024). DOI 10.1137/22M1519444
12. Dolan, E.D., Moré, J.J.: Benchmarking optimization software with performance profiles. *Mathematical Programming* **91**(2), 201–213 (2002). DOI 10.1007/s101070100263
13. Duchi, J., Hazan, E., Singer, Y.: Adaptive Subgradient Methods for Online Learning and Stochastic Optimization. *Journal of Machine Learning Research* **12**(61), 2121–2159 (2011). URL <http://jmlr.org/papers/v12/duchi11a.html>
14. Fowkes, J., Roberts, L., Bűrmen, Á.: PyCUTEst: An open source Python package of optimization test problems. *Journal of Open Source Software* **7**(78), 4377

- (2022). DOI 10.21105/joss.04377
15. Gao, W., Goldfarb, D.: Quasi-Newton Methods: Superlinear Convergence Without Line Searches for Self-Concordant Functions. arXiv preprint arXiv:1612.06965 (2018). DOI 10.48550/arXiv.1612.06965
 16. Gould, N.I., Orban, D., Toint, P.L.: CUTEst: A Constrained and Unconstrained Testing Environment with safe threads for mathematical optimization. *Comput. Optim. Appl.* **60**(3), 545–557 (2015). DOI 10.1007/s10589-014-9687-3
 17. Gower, R.M., Goldfarb, D., Richtárik, P.: Stochastic Block BFGS: Squeezing More Curvature out of Data. arXiv preprint arXiv:1603.09649 (2016). DOI 10.48550/arXiv.1603.09649
 18. Grapiglia, G.N., Stella, G.F.D.: An adaptive trust-region method without function evaluations. *Computational Optimization and Applications* **82**(1), 31–60 (2022). DOI 10.1007/s10589-022-00356-0
 19. Gratton, S., Jerad, S., Toint, Ph.L.: Complexity of a class of first-order objective-function-free optimization algorithms. *Optimization Methods and Software* **41**(2), 478–508 (2026). DOI 10.1080/10556788.2023.2296431
 20. Gratton, S., Toint, P.L.: A note on solving nonlinear optimization problems in variable precision. *Computational Optimization and Applications* **76**(3), 917–933 (2020). DOI 10.1007/s10589-020-00190-2
 21. Gratton, S., Toint, P.L.: OFFO minimization algorithms for second-order optimality and their complexity. *Computational Optimization and Applications* **84**(2), 573–607 (2023). DOI 10.1007/s10589-022-00435-2
 22. Hassan, B.A., Moghrabi, I.A.R.: A modified secant equation quasi-Newton method for unconstrained optimization. *Journal of Applied Mathematics and Computing* **69**(1), 451–464 (2023). DOI 10.1007/s12190-022-01750-x
 23. Higham, N.J.: Accuracy and Stability of Numerical Algorithms. Society for Industrial and Applied Mathematics (2002). DOI 10.1137/1.9780898718027
 24. Izycheva, A., Darulova, E.: On sound relative error bounds for floating-point arithmetic. In: Proceedings of the 17th Conference on Formal Methods in Computer-Aided Design. FMCAD Inc, Austin, Texas (2017)
 25. Kanzow, C., Steck, D.: Regularization of limited memory quasi-Newton methods for large-scale nonconvex minimization. *Mathematical Programming Computation* **15**(3), 417–444 (2023). DOI 10.1007/s12532-023-00238-4
 26. Li, D.H., Fukushima, M.: On the Global Convergence of the BFGS Method for Nonconvex Unconstrained Optimization Problems. *SIAM Journal on Optimization* **11**(4), 1054–1064 (2001). DOI 10.1137/S1052623499354242
 27. Li, Y.J., Li, D.H.: Truncated regularized Newton method for convex minimizations. *Computational Optimization and Applications* **43**(1), 119–131 (2009). DOI 10.1007/s10589-007-9128-7
 28. Liu, D.C., Nocedal, J.: On the limited memory BFGS method for large scale optimization. *Mathematical Programming* **45**(1), 503–528 (1989). DOI 10.1007/BF01589116
 29. Mannel, F.: A Globalization of L-BFGS and the Barzilai–Borwein Method for Nonconvex Unconstrained Optimization. *Journal of Optimization Theory and Applications* **204**(3), 1–34 (2025). DOI 10.1007/s10957-024-02565-5

30. Mannel, F., Om Aggrawal, H., Modersitzki, J.: A structured L-BFGS method and its application to inverse problems. *Inverse Problems* **40**(4), 045022 (2024). DOI 10.1088/1361-6420/ad2c31
31. Moré, J.J., Thunte, D.J.: Line search algorithms with guaranteed sufficient decrease. *ACM Trans. Math. Softw.* **20**(3), 286–307 (1994). DOI 10.1145/192115.192132
32. Moré, J.J., Wild, S.M.: Estimating Computational Noise. *SIAM Journal on Scientific Computing* **33**(3), 1292–1314 (2011). DOI 10.1137/100786125
33. Nesterov, Y., Polyak, B.: Cubic regularization of Newton method and its global performance. *Mathematical Programming* **108**(1), 177–205 (2006). DOI 10.1007/s10107-006-0706-8
34. Nocedal, J., Wright, S.J.: *Numerical Optimization*. Springer (1999). DOI 10.1007/978-0-387-40065-5
35. Okazaki, N.: libLBFGS: A library of limited-memory broyden-fletcher-goldfarb-shanno (l-BFGS) (2007). URL <https://github.com/chokkan/liblbfgs>
36. Paquette, C., Scheinberg, K.: A Stochastic Line Search Method with Expected Complexity Analysis. *SIAM Journal on Optimization* **30**(1), 349–376 (2020). DOI 10.1137/18M1216250
37. Ranganath, A., Singhal, M., Marcia, R.: Symmetric Rank-One Quasi-Newton Methods for Deep Learning Using Cubic Regularization. In: 2025 IEEE 10th International Workshop on Computational Advances in Multi-Sensor Adaptive Processing (CAMSAP), pp. 121–125 (2025). DOI 10.1109/CAMSAP66162.2025.11423871
38. Ruan, H., Yang, W.: Adaptive Regularized Quasi-Newton Method Using Inexact First-Order Information. *Journal of Computational Mathematics* **42**(6), 1656–1687 (2024). DOI 10.4208/jcm.2306-m2022-0279
39. Sheng, Z., Yuan, G., Cui, Z.: A new adaptive trust region algorithm for optimization problems. *Acta Mathematica Scientia* **38**(2), 479–496 (2018). DOI 10.1016/S0252-9602(18)30762-8
40. Shi, H.J.M., Xie, Y., Byrd, R., Nocedal, J.: A Noise-Tolerant Quasi-Newton Algorithm for Unconstrained Optimization. *SIAM Journal on Optimization* **32**(1), 29–55 (2022). DOI 10.1137/20M1373190
41. Sun, S., Nocedal, J.: A trust region method for noisy unconstrained optimization. *Mathematical Programming* **202**(1), 445–472 (2023). DOI 10.1007/s10107-023-01941-9
42. Ueda, K., Yamashita, N.: Convergence Properties of the Regularized Newton Method for the Unconstrained Nonconvex Optimization. *Applied Mathematics and Optimization* **62**(1), 27–46 (2010). DOI 10.1007/s00245-009-9094-9
43. Ueda, K., Yamashita, N.: A regularized Newton method without line search for unconstrained optimization. *Computational Optimization and Applications* **59**(1), 321–351 (2014). DOI 10.1007/s10589-014-9656-x
44. Virtanen, P., Gommers, R., Oliphant, T.E., Haberland, M., Reddy, T., Cournapeau, D., Burovski, E., Peterson, P., Weckesser, W., Bright, J., van der Walt, S.J., Brett, M., Wilson, J., Millman, K.J., Mayorov, N., Nelson, A.R.J., Jones, E., Kern, R., Larson, E., Carey, C.J., Polat, I., Feng, Y., Moore, E.W., VanderPlas, J., Laxalde, D., Perktold, J., Cimrman, R., Henriksen, I., Quintero, E.A., Har-

- ris, C.R., Archibald, A.M., Ribeiro, A.H., Pedregosa, F., van Mulbregt, P., SciPy 1.0 Contributors: SciPy 1.0: Fundamental algorithms for scientific computing in python. *Nature Methods* **17**, 261–272 (2020). DOI 10.1038/s41592-019-0686-2
45. Wang, S., Fadili, J., Ochs, P.: Convergence rates of regularized quasi-Newton methods without strong convexity. *arXiv preprint arXiv:2506.00521* (2025). DOI 10.48550/arXiv.2506.00521
46. Ward, R., Wu, X., Bottou, L.: AdaGrad stepsizes: Sharp convergence over non-convex landscapes. *Journal of Machine Learning Research* **21**(219), 1–30 (2020). URL <http://jmlr.org/papers/v21/18-352.html>
47. Xie, Y., Byrd, R.H., Nocedal, J.: Analysis of the BFGS Method with Errors. *SIAM Journal on Optimization* **30**(1), 182–209 (2020). DOI 10.1137/19M1240794
48. Yabe, H., Ogasawara, H., Yoshino, M.: Local and superlinear convergence of quasi-Newton methods based on modified secant conditions. *Journal of Computational and Applied Mathematics* **205**(1), 617–632 (2007). DOI 10.1016/j.cam.2006.05.018
49. Yuan, G., Wei, Z.: Convergence analysis of a modified BFGS method on convex minimizations. *Computational Optimization and Applications* **47**(2), 237–255 (2010). DOI 10.1007/s10589-008-9219-0
50. Zhang, H., Ni, Q.: A new regularized quasi-Newton method for unconstrained optimization. *Optimization Letters* **12**(7), 1639–1658 (2018). DOI 10.1007/s11590-018-1236-z
51. Zhang, J., Xu, C.: Properties and numerical performance of quasi-Newton methods with modified quasi-Newton equations. *Journal of Computational and Applied Mathematics* **137**(2), 269–278 (2001). DOI 10.1016/S0377-0427(00)00713-5
52. Zhang, J.Z., Deng, N.Y., Chen, L.H.: New Quasi-Newton Equation and Related Methods for Unconstrained Optimization. *Journal of Optimization Theory and Applications* **102**(1), 147–167 (1999). DOI 10.1023/A:1021898630001

Statements and Declarations

Funding

This work was partially supported by JSPS KAKENHI (23H03351, 24K23853) and JST CREST (JPMJCR24Q2).

Competing Interests

The authors have no relevant financial or non-financial interests to disclose.

Author Contributions

All authors contributed to the study conception and design. The first draft of the manuscript was written by Hiroki Hamaguchi, and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

Data Availability

All data analyzed during this study are publicly available. URLs are included in this article.

Code Availability

The code used in this study is publicly available at the following GitHub repository:
<https://github.com/HirokiHamaguchi/qnlab>.